Linguistic transparency as mediated accessibility

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11th May 2026

Abstract

Linguistic transparency is best understood as a family of mediated accessibility relations rather than a unitary syntactic property. A construction, expression, or grammatical configuration is transparent when information that might otherwise be encapsulated remains accessible for agreement, interpretation, constructional generalization, processing, or cross-linguistic comparison. The umbrella schema *X is transparent w.r.t. property P for relation R when P remains accessible through X for the purposes of R* is a comparative concept in Haspelmath's sense: an analyst's question, applicable to any language, that asks which configurations let an internal property reach a grammatical or interpretive relation.

The schema's job is to locate PROJECTIBILITY PROFILES: cases where an observable base licenses non-trivial projection to a target, stabilized by some identifiable source. Since projectibility is field-relative, the paper keeps the fields apart. Its primary claim is syntactic: some transparency attributions support induction over agreement, syntactic-category licensing, complement selection, and related grammatical behaviour.

CGEL's number-transparent quantificational nouns are the strongest profile worked through: a small, relatively stable class whose membership predicts agreement override, characteristic determiner patterns, partitive-vs-non-partitive *of*-complement availability, fused-head behaviour, count/non-count percolation, and premodification tendencies, stabilized by conventional lexical inheritance. Transparent free relatives supply a second syntactic profile: constructional sub-type and verb-class membership predict agreement transparency across the parent construction and syntactic-category transparency in the attributional daughter.

Later sections ask adjacent field-relative questions in construction grammar, compound semantics, morphological processing, typology, and formal semantics. The form–meaning accessibility family across constructional compositionality, compound rated transparency, and morphological priming licenses within-domain prediction in three operational windows; whether the family constitutes a cross-domain projectibility cluster on shared items is an open research question. The projectibility profile is the main concept; Slater's stable-property-cluster (SPC) framework is the general diagnostic; Khalidi's causal-network account is reserved as a possible refinement for profiles with demonstrated causal ordering, applied here only where the evidence

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motivates it. Coordination is excluded as a non-instance; referential transparency is acknowledged as the schema's boundary case.

Against the background of stable-property-cluster and causal-network accounts of kinds, the paper extends projectibility-first analysis to a relational concept; kindhood is a downstream verdict on profiles that earn it, not the master question of the analysis.

1 Introduction

Two examples from Huddleston and Pullum's *Cambridge Grammar of the English Language* (CGEL) make the puzzle visible (2002, §18.2). *A number of spots have/*has appeared. Heaps of money has/*have been spent.* In the first sentence, the surface head *number* is singular, but agreement is plural because the *of*-complement *spots* is plural count. In the second, the surface head *heaps* is plural, but agreement is singular because the *of*-complement *money* is non-count. The construction lets the complement's number/count profile reach the verb across mediators with opposite morphological number.

Number-transparent quantificational nouns is CGEL's term. Core members include *lot, plenty, lots, bags, heaps, loads, oodles, stacks, remainder, rest, number,* and *couple*, with *majority* and *minority* at the boundary. Membership is small and relatively stable, and members head NPs whose agreement override is, on CGEL's analysis, the default pattern, contrasting with the optional override of ordinary collective nouns. Many other configurations behave similarly: a form mediates between an internal property and a grammatical or interpretive relation, and the property remains accessible through the mediator. The umbrella schema for the family follows.

> X is *transparent* with respect to property P for relation R when P remains accessible through X for the purposes of R.

The schema is a tool for a particular question: when is calling something *transparent* doing analytical work, and when is it just a label? Sometimes the label supports projection; sometimes it doesn't. The number-transparent quantificational noun construction shows the projective case.

The schema's substantive content is the *remains accessible through* X clause. A case fits only if X has internal structure whose default behaviour would have made P inaccessible to R. Cases without that default-conceal background are excluded. Coordination (§2.3) is the textbook exclusion: *Bob and Jane are happy* has no head whose morphological number competes for agreement with the conjuncts' features, so the construction has nothing to hide, and calling coordination *transparent* would be a relabel rather than an analysis. Everyday senses of transparent prose or transparent intentions, where no specific mediator-property-relation triple is in play, are excluded for a related reason: the schema's slots have no values to bind. The exclusion mechanism is what keeps the schema from cataloguing every use of *transparent* in linguistics, and the rest of the paper is occupied with cases where the slots have non-trivial values.

Once we recognize that *lot, plenty, lots, bags, heaps, loads, oodles, stacks, remainder, rest, number,* and *couple* (in its quantificational sense) form a small, conventionally stable class, we can predict more than the agreement override that motivated the label. Members of the class show characteristic patterns of determiner co-occurrence; they admit partitive *of*-complements in some subgroups but not others; they behave distinctively under the fused-head construction; the count-or-non-count profile of the *of*-complement determines the profile of the whole NP; modification by ordinary adjectives is restricted in characteristic ways. Knowing that a noun belongs to the class lets us predict much of the rest, with the predictions reliable for the core members and graded at the boundary (*majority, minority*, sense-split *couple*). The label is doing work: a small lexical inventory whose membership predicts a cluster of correlated behaviours, attenuated at the edges rather than guaranteed everywhere.

Cases like this are PROJECTIBILITY PROFILES. A profile has three components: an OBSERVABLE INPUT (here, lexical-class membership); a set of *predictions* the input licenses (here, the cluster of agreement, distributional, and modificational behaviours); and a STABILIZING PROCESS that holds the input together (here, ordinary lexical inheritance). The number-transparent class is a strong example. The predictions are non-trivial; the inventory is stable; the stabilizing process is the same one speakers use for any closed grammatical class.

Not every transparency label names a profile this strong. Some name profiles whose predictions are weaker, or whose stabilizing process is different, or both. The paper asks which transparency labels do projection-supporting work, and what makes them able to do it.

Projectibility is field-relative: a profile supports induction for a field's explanatory and descriptive purposes, not for every inquiry at once. The primary field here is syntax. The central question is whether a transparency attribution lets syntacticians project from an observable lexical or constructional base to agreement, syntactic-category licensing, complement selection, and other grammatical behaviours. The answer is strongest for number-transparent quantificational nouns and transparent free relatives.

Later sections ask neighbouring field-relative questions. Construction grammar asks about interpretation, learning, and analogical generalization; compound semantics and corpus modelling ask about rated transparency; psycholinguistic processing asks about priming; typology asks about cross-linguistic comparison; formal semantics supplies a boundary case. Each field has its own input, targets, stabilizer, and evidential standard.

X (the mediator) can be a lexical item, a partly or fully schematic construction, an intensional context, or a derivationally complex word. P can be a number/count feature, a constituent's meaning, a referent, or a stem's lexical representation. R can be agreement, interpretation, substitution under co-reference, or masked priming. Accessibility itself admits two readings: class-like (core members behave categorically, even where class boundaries are fuzzy) and gradient (compositionality scales, transparency ratings, priming magnitudes).

The argument is constructive rather than exhaustive. The paper leaves aside most uses of TRANSPARENCY in linguistics, as well as extraction, islands, barriers, and phases; extraction-transparency is acknowledged in §2 but not developed. It also leaves aside phonological or orthographic transparency and the everyday senses of *transparent explanation* or *transparent terminology*. Referential or DE RE/DE DICTO transparency receives a controlled mention in §4 as the boundary case where the schema's mediator-form structure stretches thinnest.

MEDIATED ACCESSIBILITY is more specific than CLARITY and broader than COMPOSITIONALITY. Clarity is rhetorical: it doesn't single out a mediator or a relation. Compositionality is the limiting case of FORM–MEANING transparency, where a mediator's parts predict the whole. The umbrella schema is broader: it covers form–meaning cases (constructions, compounds, derived words) and accessibility cases that aren't reducible to whether parts predict wholes. Number transparency is the paradigm of the second type: a quantificational noun's morphological number doesn't by itself predict the verb's number, and under the construction it doesn't have to.

Section 2 takes up morphosyntactic transparency, with number-transparent quantificational nouns and transparent free relatives as the paradigm cases. Sections 3 to 5 work through the form–meaning accessibility family: theoretical compositionality (Goldberg, 2006), rated compound transparency (Bell & Schäfer, 2016), and time-sensitive lexical processing (Heyer & Kornishova, 2018). Section 6 applies Haspelmath's (2010) comparative-concept discipline to the schema's cross-linguistic ambitions. Section 7 names the stabilizing structures behind each profile and treats Slater's SPC framework (Slater, 2015) and Khalidi's causal-network account (Khalidi, 2018) as complementary diagnostics for projection-supporting stability.

2 Morphosyntactic transparency

Morphosyntactic transparency is the case where X is a syntactic construction (or a lexical class within one) and R is a matching relation between a non-head property and a co-occurring exponent.[1] Subject-verb agreement is the textbook instance. CGEL's number-transparent quantificational nouns are the paradigm; collective NPs are a transitional case; coordination is a non-instance; transparent free relatives add a second profile, stabilized by constructional rather than lexical inheritance, with two dissociable R-relations under one mediator. Section 2.6 marks the Korean extension as deferred rather than treating it as evidence for the present syntactic result.

This section carries the paper's primary field-relative claim. The projected targets are syntactic and morphosyntactic: agreement, syntactic-category licensing, complement selection, fused-head behaviour, and distributional restrictions inside NP and fused-relative constructions. Nothing in the syntactic result depends on phrase-structure representation as such: the profiles require only observable relations among a mediator, an internal property, and the grammatical relation that tracks it, whether those relations are represented in constituency or dependency terms.

2.1 Number-transparent quantificational nouns

The paradigm again is CGEL's two-direction contrast: *A number of spots have/*has appeared* (plural agreement despite singular *number*) and *Heaps of money has/*have been spent* (singular agreement despite plural *heaps*). Schema-wise, X is the number-transparent quantificational noun + *of*-complement construction; P is the number/count profile of the *of*-complement; R is subject-verb agreement.

CGEL's "simple agreement rule" is head-driven: a singular head NP triggers singular agreement, a plural head NP plural agreement (Huddleston & Pullum, 2002, §18.1). Number-transparent NPs override the rule. CGEL describes the override as obligatory in the central cases; rare singular-verb attestations with *number* are noted in footnote 73 to §18.2 and analyzed there as hypercorrections rather than as variant grammar. The contrast with the optional override of collective nouns (§2.2) is

[1] Extraction transparency (whether an island or barrier is opaque or transparent to long-distance dependency formation) is a morphosyntactic-accessibility relation under the schema (with R = dependency formation), and the associated question remains active. Transformational accounts posit empty elements (traces, copies) that mediate the connection between a fronted element and its base position; non-transformational dependency accounts (da Cunha & Gibson, 2025; Gibson, 2025) argue that no such mediating empty elements are needed and that direct head-to-dependent connections better fit acceptability and processing data. Both sides make schema-level claims about whether X contains invisible mediating structure. The centre of the present argument is the agreement-style cases.

the operative point. The construction's job is to let the *of*-complement's number/count profile reach agreement past the head's morphological number, which would otherwise have determined it.

The core inventory is small and well documented. CGEL identifies the principal members in example [57]: *lot, plenty* (singular forms; *lot* most often with *a*-determiner, *plenty* typically without a determiner); *lots, bags, heaps, loads, oodles, stacks* (plural forms, typically without a determiner); *remainder, rest* (singular forms typically with *the*-determiner); *number, couple*$_2$ (singular forms typically with *a*-determiner, taking only plural *of*-complements, partitive or non-partitive). Of these, "the clear cases of number-transparent singular nouns are *lot, number*, and *couple*$_2$"; *majority* and *minority* are borderline (CGEL [13] notes plural override is "surely obligatory" in some examples but singular agreement is occasionally attested) (Huddleston & Pullum, 2002, §18.2). The determiner patterns above are typical, not absolute: *the lot of it* (in a totality sense) is grammatical alongside *a lot of it*, and similar sense-conditioned variations exist for other members.

Class membership predicts a cluster of behaviours beyond the agreement override. The relevant contrast is between PARTITIVE *of*-complements, which require a definite complement that picks out a subset of a definite set (*of the money*, *of those books*, *of these soldiers*), and NON-PARTITIVE *of*-complements, which take indefinite complements (*of money*, *of soldiers*, *of some paintings*). *Lot* most often takes *a* as determiner with limited premodification (*a whole/huge lot of time*) (Huddleston & Pullum, 2002, §3.3); *the lot of it* is also available in a totality sense, but other determiners are blocked (**some lot of it*). *Plenty* typically takes no determiner or modifier (*plenty of butter*, not generally *a remarkable plenty of butter*), and like the plural-only forms it allows virtually no partitive *of*-complements.[2] The plural-only forms (*lots, bags, heaps, loads, oodles, stacks*) show the same pattern: *They have oodles of money* is normal; *Oodles of the money had already been spent* is rare to non-attested.

Remainder and *rest* take only partitive *of*-complements in their characteristic use: *the remainder of the time*, not *the remainder of time*. *Number* and *couple*$_2$ take plural obliques: *a number of the protesters*, not *a number of money*. The pattern is highly structured even where individual cases admit sense-conditioned variation.

The pattern shows what number transparency *is* in the umbrella schema's vocabulary. X (the construction) leaves P (the number/count profile of the *of*-complement) accessible to R (agreement). What X conceals from agreement is the morphological number of the head noun *lot, plenty, heaps*, etc. The agreement-targeting feature is on the *of*-complement, not on the surface head. That's what makes X transparent with respect to P: P is on the complement, not on the head, and X lets P reach the verb anyway.

2.2 Collective NPs as a transitional case

Singular collective nouns (*committee, jury, family, team, government*; also *clergy, intelligentsia, public*, etc.) allow optional plural override of head-driven agreement: *The committee is/are divided* (CGEL [8]) (Huddleston & Pullum, 2002, §18.2). The override is meaning-sensitive: singular agreement focuses on the unit, plural on the members. Plural is unavailable with predicates that apply to the

[2] A pilot COCA check (May 2026) across matched complements quantifies the contrast: plural-only forms show $\approx 0.2\%$ partitive uses; *plenty* $\approx 0.03\%$; *rest* / *remainder* show the reverse pattern at $\geq 97\%$ partitive (raw, higher after filtering idiomatic *for the rest of time*). Plural-only forms also reject *of* [*a* N] (0 hits across tested complements), indicating their *of*-complement is bare/non-determined rather than just non-partitive. Full counts and methodology in Appendix A.2; raw data in `paper/data/coca-pilot.md`.

whole but not individual members (*The committee consists/*consist of two staff and three students*) and is "of questionable acceptability" if the collective takes determiner *one* or *another* (*One committee, appointed last year, has/?have not yet met*).

Both collective NPs and number-transparent quantificational NPs override the head's morphological singular at agreement. The override is possible at all because plural agreement requires only non-singular construal, not the precise atomic enumeration that singular *a*(*n*) and low cardinals demand. Individuation, the speaker's construal of referents as discrete units and the variable that count grammar tracks, comes in different intensities. Loose count properties (agreement, *many/few*, plural-only forms) can be satisfied by individuation alone; tight properties (singular *a*(*n*), low cardinals) can't, since they require selecting exactly one atom or enumerating a precise *n*-tuple. That asymmetry is why plural agreement can override a morphologically singular head while tight properties on the same NP follow ordinary count-noun behaviour. The override is selective in this way: it operates at the loose end of the cluster, not across it. Traditional grammar calls the agreement-side reflex notional concord. Reynolds (2026a) develops the loose/tight asymmetry as a precision-driven dissociation pattern across English count nouns and proposes a bidirectional link between morphosyntax and individuation as the underlying mechanism, where each cues the other and the override is the case where individuation outruns the head's morphological signal.

Where the non-singular construal is sourced differs in the two cases. In collectives, individuation is lexically encoded in the head: a singular collective denotes a unit whose members remain individuable, and the agreement relation accesses that member-level construal under predicates that license it. In number-transparent quantificational NPs, individuation is supplied by the *of*-complement: *a lot of tourists* sources non-singular construal from *tourists*; *a lot of money* sources low (non-count) construal from *money*; the construction projects the complement's count/non-count profile to agreement. The former is lexical-semantic accessibility of member individuation; the latter is constructional accessibility of the *of*-complement's individuation profile.

The two sources also differ in how stably the construal projects. The constructional source projects uniformly: in number-transparent QNs the override applies regardless of predicate type. The lexical-semantic source projects selectively: collective override requires the predicate to license member-level construal, which is why *The committee consists/*consist of two staff and three students* blocks the override even with the same singular collective head, and why singular-only determiners degrade plural override (*One committee, appointed last year, has/?have not yet met*).

Both routes are mediated accessibility under the umbrella schema; the difference is whether the individuation profile is baked into the head's lexical semantics or imported from the *of*-complement.

CGEL itself notes that "the division between collective and number-transparent nouns is by no means sharply drawn" (Huddleston & Pullum, 2002, §18.2). Some nouns straddle. *Couple* has two senses: *couple*$_1$ (a man and woman in a relationship) is a regular collective; *couple*$_2$ ("approximately two") is number-transparent. *Bunch* tilts collective for inanimate aggregates (*A bunch of flowers was presented to the teacher*) but transparent for groups of people (*A bunch of hooligans were seen leaving the premises*) (Huddleston & Pullum, 2002, §18.2).[3]

[3] A pilot COCA check (May 2026) supports the animate half (40 plural / 0 singular for *a bunch of people / kids*) but fails to support the inanimate half: *a bunch of flowers was* and the CGEL example *A bunch of flowers was presented to the teacher* return 0 hits, and the two inanimate-bunch hits in subject position that did surface take plural rather than singular. The COCA evidence treats *bunch*-as-subject as transparent for animate aggregates and the inanimate-singular

The schema accommodates both the paradigm and the transitional case, but they have different internal structure. Number-transparent NPs are class-like, with the override the default behaviour for class members. Collective NPs are gradient: the override probability depends on the head, the predicate, the determiner, the dialect register, and the construal the speaker has in mind. A theory of morphosyntactic transparency that ignored the difference between a class-based default override and a construal-driven optional one would lose the interesting structure of both cases.

2.3 Coordination as a non-instance

Coordination of singulars triggers plural agreement: *Bob and Jane are happy*. This looks like a transparency case at first (the conjuncts' number features percolate up to the coordinate construction and determine agreement), but the schema excludes it. The schema requires that X *could have* made P inaccessible. The coordinate construction does no such hiding: there's no head whose morphological number competes with the conjuncts' features for agreement. The plural agreement is compositional addition (1 + 1 = 2), not mediation. Some coordinate NPs take singular agreement: *Bacon and eggs is my favourite breakfast*; *Wine and cheese is expected*; *Fish and chips is expensive these days*. These leave the exclusion intact. In such cases agreement tracks a holistic construal of the coordinate phrase as a dish, package, event type, or conventional pairing. The singular feature isn't a property of either conjunct that remains accessible through the coordinate construction; it's a construal of the coordinate phrase as a whole. Ordinary plural coordination and holistic singular coordination therefore differ in agreement semantics, but neither is transparent in the schema's sense. The former is compositional summation; the latter is whole-phrase unit construal.[4] CGEL treats coordination-and-agreement under a separate heading (Huddleston & Pullum, 2002, §18.4), not alongside the override constructions of §18.2.

The clean distinction across §2's cases: in collective plural agreement, internal member plurality becomes accessible through a singular head; in number-transparent QNs, complement number becomes accessible through a quantificational head; in unit-construal coordination, the whole coordinate phrase is construed as singular and no internal property is being made transparent. The first two are mediated accessibility; the third isn't.

The "X could have made P inaccessible" clause works as an independent diagnostic, not an ad hoc filter. The test asks whether the construction-type admits a default-conceal variant: is there a version of the same construction-type, attested in the language, in which X's internal structure makes P inaccessible to R? For English NPs, head-driven agreement is the default and the number-transparent class is the marked exception within the same NP type. For coordinate constructions no such variant exists: the type admits no version in which conjunct features fail to reach agreement, so coordination has no default-conceal backdrop. The diagnostic distinguishes mediator-types from compositional-summation types: transparency is informative only where the same mediator type has opaque or partially opaque alternatives.

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pattern as an analyst's example rather than a corpus regularity. Full counts: Appendix A.4.

[4] Cases like *One and one is two* are an arithmetic-expression construction, not a meal/package coordination: *and* functions close to an addition operator, and the subject denotes an operation, equation, or expression. Not transparency either.

2.4 Number-transparent NPs as a projectibility profile

Section 1 set the diagnostic question: does identifying a candidate transparency profile let us predict behaviour? For number-transparent quantificational nouns the answer is yes, with the prediction running in two task settings.

For known CGEL members (*lot, plenty, lots, bags, heaps, loads, oodles, stacks, remainder, rest, number, couple*$_2$), lexical identity is sufficient input: knowing a noun is on this list licenses projection to the rest of the profile.

For candidate new members, class discovery uses two diagnostics independently observable from any of the projected behaviours: quantificational semantics (an indefinite quantity rather than a precise count or kind) and high-frequency occurrence as head of an *of*-complement construction. A candidate that satisfies both is then tested against the projected cluster:

- near-obligatory agreement override in the central cases, singular- or plural-direction depending on subgroup;[5]
- characteristic determiner patterns by subgroup (*lot* most often with *a*, also with *the* in totality uses; *plenty* typically without a determiner; the plural-only forms typically without a determiner; *remainder, rest* typically with *the*; *number, couple*$_2$ typically with *a*);
- partitive vs. non-partitive *of*-complement availability;
- behaviour under the fused-head construction;
- count/non-count percolation of the *of*-complement to the whole NP, with consequences for selection by predicates that are count- or non-count-sensitive;
- restricted premodification (*a whole/huge lot* but not *a remarkable plenty*).

The class's sub-group structure makes the cluster's predictions concrete. Table 1 displays the per-noun pattern for the central diagnostics; the entries are CGEL-coded rather than corpus-coded, and the within-domain audit a fuller study would supply is left as the within-domain check §2.4's falsification condition references.

Table 1. Number-transparent quantificational nouns: behavioural matrix (CGEL-based pilot).

Sub-group / Noun	Determiner	Partitive *of* N	Non-partitive *of* N	Override direction
Singular, *a*-determined				
lot	*a*; also *the* (totality)	✓ (*a lot of the money*)	✓ (*a lot of money*)	sg → pl with plural complement
number	*a*	✓ (plural oblique)	✓ (plural oblique)	sg → pl with plural complement

[5] A pilot COCA check (May 2026) supports the near-categorical pattern: predicted-direction agreement runs 98–100% across nine tested QN+complement combinations after KWIC filtering of checked counter-direction cells. *A lot of money* required parse-shift filtering (10 of 13 raw plural-agreement hits were inside relative clauses modifying a plural head, e.g. *people who earn a lot of money are successful*; full filtering in Appendix A.3). The large *a lot of people is/was* counter-direction cell remains unfiltered, so the 98.0% row is conservative. Full counts and methodology: Appendix A.3.

Sub-group / Noun	Determiner	Partitive *of* N	Non-partitive *of* N	Override direction
couple$_2$	*a*	✓ (plural oblique)	✓ (plural oblique)	sg → pl with plural complement
Singular, usually without a determiner				
plenty	usually without a determiner	not typical	✓ (*plenty of money*)	sg → pl with plural complement
Plural, usually without a determiner				
lots, bags, heaps, loads, oodles, stacks	usually without a determiner	× (COCA pilot: ≈0.2%, §2.1)	✓	pl → sg with non-count complement
Singular, *the*-determined				
remainder	*the*	✓ (*the remainder of the time*)	× (**the remainder of time*)	sg (default)
rest	*the*	✓ (*the rest of the time*)	× (**the rest of time*)	sg (default)
Boundary				
majority, minority	*a*	✓	?	variable
bunch	*a*	✓	✓	variable

Override applies in the central cases for all class members; boundary cases show variable behaviour (§2.2). Fused-head behaviour and count/non-count percolation are predicted to apply across class members but aren't per-noun-audited here. Premodification is restricted across the class but with sub-group-specific patterns (*a whole/huge lot*; minimal modification with *plenty*; fewer attested premodifiers with the plural-only forms).

Each sub-group covaries: knowing a candidate noun's determiner pattern and partitive availability narrows the sub-group, which then licenses the override-direction prediction (and vice versa). The boundary cases have weaker covariance, which is why §2.2's gradient framing applies to them.

Figure 1 visualizes the override-direction data from the COCA pilot. Predicted-direction agreement runs 98–100% across nine tested QN+complement combinations after KWIC filtering of checked counter-direction cells. The override is essentially categorical; small-*n* cells (*plenty of money*, *the rest of the money*, *the rest of the people*) have wide confidence intervals even at 100% point estimates.

This two-task structure keeps the prediction genuinely projective for the new-member case: in-

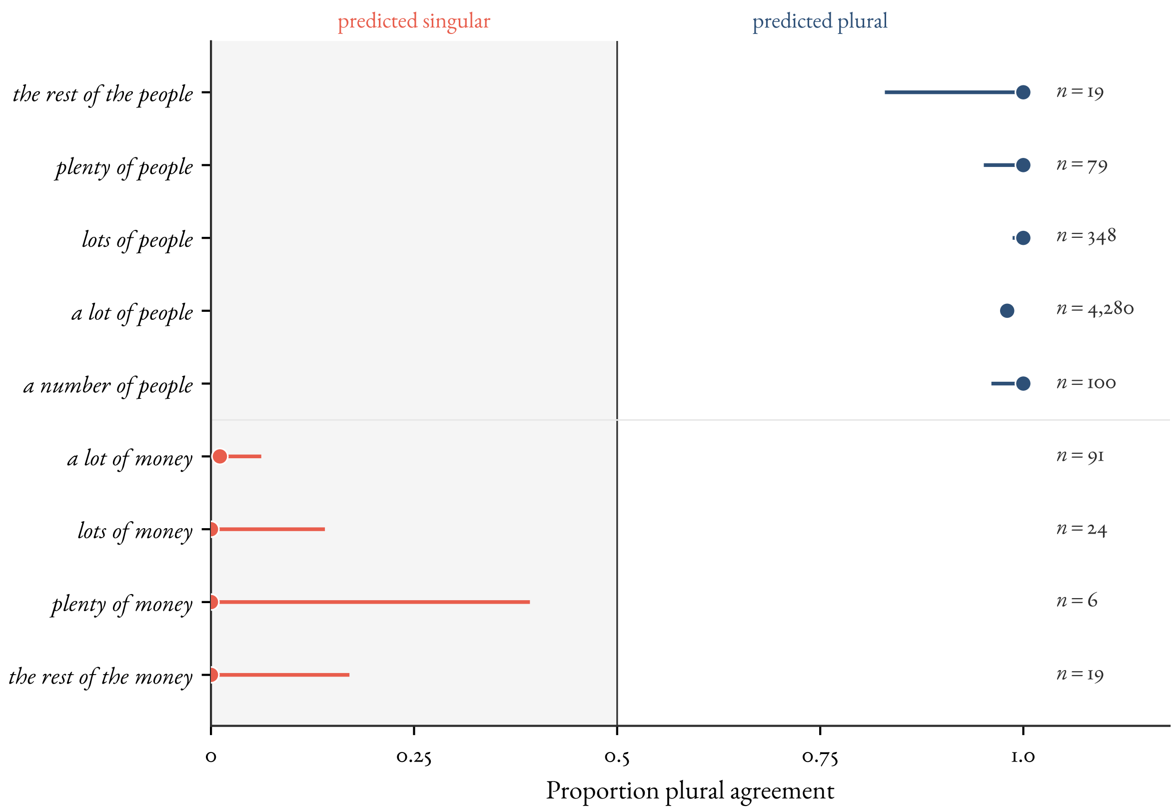

Figure 1: Override-direction agreement for nine number-transparent QN + complement combinations in COCA. Each row is a QN+complement; the x-axis is the proportion of finite-verb agreement in the plural direction (0 = all singular, 1 = all plural). Predicted-singular cases (coral) cluster at the left; predicted-plural cases (blue) cluster at the right. Horizontal bars are 95% Wilson binomial confidence intervals; *n* is the total finite-verb count for that subject after KWIC filtering where available. See footnote and `paper/data/coca-pilot.md`. Data: COCA pilot, May 2026.

puts are quantificational semantics and *of*-complement-head distribution; the cluster of agreement-override / determiner-frame / partitive / fused-head / count-or-non-count / premodification behaviours is then testable on the candidate. Boundary cases like *majority* and *minority* (§2.2) and the dual-sense *couple*$_1$ (collective) versus *couple*$_2$ (number-transparent) show graded membership at the edges; the predictions hold in attenuated form, not as crisp class-conditional facts. What distinguishes prediction from redescription: the cluster supports inductive extension to *new* members (boundary cases like *majority/minority*; gradient cases like *bunch* in §2.2) that aren't built into the cluster's defining criteria.

The more demanding test is what the profile lets us predict beyond the in-grammar cluster. Three off-list projections are plausible candidates. *Diachronic*: members should enter the class through grammaticalization or sense-split rather than through arbitrary lexical addition, with recent boundary additions (*majority, minority*) and the sense-split in *couple* matching that pattern. *Processing*: if the override is grammaticalized rather than improvised, sentences with number-transparent QNs should carry no comprehension cost relative to ordinary plural-subject sentences in adult speakers; no-cost or low-cost override is the prediction, not a parsing penalty for the surface mismatch. *Cross-linguistic*: the comparative concept of mediated accessibility (§6) licenses asking whether languages with analogous partitive-quantificational configurations exhibit comparable cluster structure (French *beaucoup de*, Spanish *un montón de*); the descriptive categories needn't be equated for the cluster shape to be compared. None of these is tested in the present paper. Their availability as predictions, rather than restatements of the input criteria, is what supports treating the class as a projectibility profile in §7.1's stronger sense: a maintained-and-projectible cluster rather than just a distributional grouping.

The cluster of behaviours listed above makes number-transparent quantificational nouns the strongest candidate PROJECTIBILITY PROFILE examined here: an observable base (lexical class membership for known members; quantificational semantics plus *of*-complement distribution for candidates) supporting projection to a structured set of targets, stabilized by conventional lexical inheritance. The profile is a candidate to pass Slater's (2015) stability-plus-projection diagnostic; the within-domain check beyond override direction is empirically open. The falsification condition is concrete: if a noun satisfying the input diagnostics failed to show the predicted cluster, the profile would fail. Boundary cases (*majority*, *minority*, sense-split *couple*) reflect the conventional inheritance that stabilizes the profile. The count/non-count percolation behaviour the profile licenses participates in the broader count cluster that Reynolds (2026a) analyzes as Khalidian, with conventional individuation as a central causal node; §7.4 develops that connection. Whether the profile earns the local-kind label as a downstream verdict is for §7.

2.5 Transparent free relatives

A second morphosyntactic-transparency case fits the schema. Transparent free relatives (TFRs) are a sub-type of fused-relative construction (Huddleston & Pullum, 2002, ch. 12, §6.3) in which the matrix accesses properties of an embedded predicate (the NUCLEUS) through *what*: *what they call ecstatic*, *what we regard as essential*, *what seems to be a tourist*. *What* is a fused-relative pronoun, morphologically and syntactically a singular indefinite; the matrix accesses features of the nucleus rather than of *what*. The construction has been examined extensively in the formal-syntactic literature, with three main competing analyses: a parenthetical-with-backward-deletion treatment (Schelfhout et al., 2003; Wilder, 1999), a unified analysis on which standard and transparent free relatives share a configurational structure (Grosu, 2003, 2016), and a graft analysis in which a single constituent is shared between matrix and relative clauses (van Riemsdijk, 2006). The present treatment follows Kim's (2026, ch. 6.8) construction-grammar analysis and draws on the corpus evidence summarized in Kim and Reynolds (2026).

Two R-relations dissociate empirically, and they have different verb-class profiles. AGREEMENT TRANSPARENCY matches matrix-verb agreement to the number profile of the nucleus and is licensed across the full inventory of TFR-licensing verbs, including the *seem/appear* class: *what appear to be trousers are on the chair* (plural agreement on plural nucleus *trousers* despite singular *what*; corpus-attested form, see (Kim & Reynolds, 2026)). Attributional verbs license the same: *what they call essential workers are exhausted*. SYNTACTIC-CATEGORY TRANSPARENCY is the narrower relation: the matrix's syntactic-category requirement is mediated through the syntactic category of the nucleus, so a non-NP nucleus licenses the TFR in a non-NP slot. *He was what they call dimwitted* has an AP nucleus (*dimwitted*) and the TFR in predicative-AP position. This narrower relation is licensed only by the attributional sub-group of TFR-licensing verbs (*call, consider, describe as, regard as, take to be*); the *seem/appear* class shows agreement transparency only.[6]

A working corpus check supports the asymmetry. AP nuclei are productive with attributional verbs (estimated hundreds of hits in COCA for *call* and *consider* alone, with sampled verification) but absent with *seem* and *appear* in a 178-hit exhaustive search; *take to be* patterns with the attributional class despite carrying an overt copula, indicating the constraint is on argument structure (the predicative complement's subject identified with the matrix object) rather than the presence of *be* (Kim & Reynolds, 2026). Within the attributional class, PP nuclei are restricted to characterizing PPs (*what she'd call in love with Sam*), not locatives, suggesting a further sub-type stratification on the nucleus's predicative profile.

The projectibility analysis runs in two task settings, parallel to §2.4. For KNOWN TFR-licensing verbs, the verb's argument-structure type (subject identified with matrix subject vs. matrix object) is sufficient input: object-orientation is the diagnostic for the attributional daughter and predicts syntactic-category transparency in addition to the agreement transparency licensed by the parent. For CANDIDATE verbs whose TFR-licensing status is uncertain (*prove to be*, *turn out to be*, *happen to be* are open cases), the same argument-structure diagnostic plus the AP-nucleus test isolates daughter membership without verb-by-verb stipulation. Two off-list projections are available. DIACHRONIC: the attributional class should show stable membership over time rather than free recruitment from the broader characterizing-verb inventory, with *take to be*'s occupying a small grammaticalization pathway as the kind of marginal addition the cluster predicts. CROSS-LINGUISTIC: languages with parallel matrix-licensing constructions should exhibit comparable parent/daughter splits if the constraint is on argument-structure configuration rather than on language-particular lexical content. Neither is tested in the present paper. Their availability as predictions, beyond the input criteria, supports treating the daughter as a candidate projectibility profile rather than as a distributional grouping.

The verb-class restriction itself admits two readings, both of which predict the asymmetry; distinguishing them is a research question the present paper leaves open. On a METALINGUISTIC reading, the attributional construction is quotative: *call* and *consider* attribute a label rather than directly predicating, and the attribution context relaxes the matrix's selectional pressure on the nucleus's syntactic

[6] A reader scanning CGEL for fused-relative examples with *seem* or *appear* will find NP-nucleus cases, since CGEL ch. 12 §6.3 treats fused relatives generally rather than the TFR sub-type; those examples are agreement-transparency cases under the present analysis, consistent with the verb-class restriction holding only for syntactic-category transparency.

category in a way unavailable to ordinary predication. On a MENTION-MEDIATED reading, the nucleus is treated as a use-mention hybrid; the construction marks the nucleus as a citation form whose syntactic-category requirement is suspended. The two differ on what licenses the relaxation: speaker-attribution versus citation-marking. They make the same first-pass predictions; cancellability and projection tests on the resulting CIs would be diagnostic.

TFRs are a second morphosyntactic projectibility profile, with the cluster of agreement, syntactic-category, and verb-class facts running off constructional sub-type membership rather than off the lexical-class membership that stabilizes §2.1. The stabilizing structure is constructional inheritance with a verb-class restriction at the daughter level. The constructional analysis is taken up in §3.5; full empirical and analytical development is in joint work with Jong-Bok Kim, in preparation.

2.6 Korean as a deferred extension

Korean is relevant to the paper's cross-linguistic ambitions, but it doesn't supply evidence for the present syntactic result. The English argument rests on number-transparent quantificational nouns and transparent free relatives. Korean case-stacking and scrambling are better treated as a separate extension because their descriptive categories, diagnostics, and projected relations have to be Korean-particular.

The candidate Korean cases would bind the schema's variables differently from English. In a case-stacking analysis, X might be a layered case-marking configuration, P an argument-role or information-structural property, and R role recovery or discourse licensing. In a scrambling analysis, X might be a word-order configuration, P scope or information-structural status, and R interpretation or discourse update. Those bindings are research questions, not premises of the present paper.

The point of naming Korean here is methodological. The umbrella schema gives the analyst a comparative question to ask without assuming that English number-transparent NPs, English TFRs, Korean case-stacking, and Korean scrambling instantiate one universal descriptive category. If the Korean cases fail to support projectible profiles, that would be a negative result for those profiles, not a problem for the English syntax cases.

2.7 Bridge

The cases in Section 2 share a relation: morphosyntactic matching mediated by a construction or lexical class. Sections 3–5 shift the relation and ask adjacent field-relative questions. Form–meaning accessibility takes interpretation, rated transparency, and processing as R instead of agreement.

3 Constructional transparency

Constructions vary in how much of their internal form–meaning organization remains accessible to speakers, learners, and analysts. This is the first of three sections on the form–meaning accessibility family: §3 sets up the theoretical compositionality gradient and the central claim that constructional transparency is a network property; §4 adds rated compound transparency; §5 adds time-sensitive lexical processing. The three are within-domain profiles distinct in items and methods; whether they cohere into a cross-domain projectibility cluster is for §7.

3.1 Constructions as form–meaning pairings

Construction Grammar (henceforth CxG) treats constructions, learned form–meaning pairings, as the basic units of grammar (Goldberg, 2006). Constructions exist at all levels: morphemes (*pre-*, *-ing*);

words (*avocado, anaconda, and*); complex words (*daredevil, shoo-in*); the regular-plural pattern [N-*s*]; filled idioms (*going great guns, give the Devil his due*); partly-filled idioms (*jog* ⟨someone's⟩ *memory*, *send* ⟨someone⟩ *to the cleaners*); Covariational Conditional (*The* X-*er the* Y-*er*, e.g., *the more you think about it, the less you understand*); Ditransitive (Subj V Obj1 Obj2); Passive (Goldberg, 2006, 5, Table 1.1). The list is heterogeneous; the unifying claim is that grammatical knowledge consists of these pairings, not of derivations applied to abstract underlying structures.

Argument structure isn't provided by verbs alone but by argument-structure constructions that combine with verbs. Goldberg's worked example uses *slice* across five argument-structure constructions: transitive (*He sliced the bread*), caused motion (*Pat sliced the carrots into the salad*), ditransitive (*Pat sliced Chris a piece of pie*), way construction (*Emeril sliced and diced his way to stardom*), resultative (*Pat sliced the box open*) (Goldberg, 2006, pp. 6–7). *Slice* contributes a constant lexical meaning across all five. The argument-structure semantics (someone acting on something; causing something to move; intending someone to receive something; moving despite obstacles; causing change of state) comes from the constructions.

Two further commitments follow. CxG is USAGE-BASED: patterns are stored as constructions if some aspect is non-predictable, or if they're fully predictable but sufficiently frequent (Goldberg, 2006, p. 5). The framework is non-derivational: "what you see is what you get," with no underlying levels, no movement, and no phonologically null elements (Goldberg, 2006, p. 11). Both commitments matter for the transparency question, since they determine what counts as a construction's "parts" and what counts as the construction's contribution.

In the umbrella schema's vocabulary: X (the mediator) is the construction; P (what remains accessible) is the meanings of the parts and their compositional contribution to the whole; R (the relation) is interpretation, learning, and analogical generalization. The relation is broader than §2's agreement, and the accessibility is gradient rather than class-like.

3.2 The compositionality gradient

Compositionality is best treated not as a binary property but as a gradient. The working scale below discretizes that gradient into four points for expository convenience. CxG itself presents compositionality as continuous; the four-point scheme is borrowed from the broader idiom-classification tradition (which distinguishes fully compositional from partly schematic and idiomatic constructions), not from CxG's own architecture, and is used here as a presentation aid rather than a CxG-internal claim.

Type	Transparency profile
Fully compositional pattern	parts and arrangement predict the interpretation
Partly schematic construction	construction contributes meaning, but parts remain interpretable
Idiomatically constrained construction	construction-specific meaning partly overrides compositional access
Frozen idiom	parts have little current interpretive accessibility

At the FULLY COMPOSITIONAL end, simple transitive sentences (*Pat sliced the bread*) have parts whose meanings predict the whole. The construction contributes argument linking, tense, and agreement, but the propositional content reduces to verb-meaning plus arguments. Fully compositional cases are the limiting endpoint of form–meaning transparency, not the paradigm. The transitive construction-type admits a default-conceal variant (frozen idioms like *kick the bucket* conceal compositional access), so any transitive sentence sits on a gradient where concealment is a real option for the construction-type; fully compositional sentences register as transparency at the trivial limit of that gradient. This is what differentiates them from §2.3's coordination case: the coordinate construction-type admits no default-conceal variant, so coordination doesn't sit on the gradient at all and the schema excludes it.

Partly schematic constructions are where CxG does most of its empirical work. The Ditransitive (Subj V Obj1 Obj2) is the canonical example. *Liza bought Zach a book* and *Liza sent Stan a book* both inherit the construction's meaning of intended transfer to an animate recipient, even though *buy* and *send* don't lexically encode that meaning. The animacy constraint on the recipient explains the contrast between *Liza sent Stan a book* and ??*Liza sent storage a book* (Goldberg, 2006, p. 10). The construction contributes transfer-with-animate-recipient meaning; the verb contributes its lexical content. The two are interpretable independently and combine compositionally, but the combination is constructional, not just lexical.

Idiomatically constrained constructions include partly-filled idioms (*jog* ⟨someone's⟩ *memory*; *send* ⟨someone⟩ *to the cleaners*). The construction has a fixed lexical core whose meaning is metaphorical or non-literal, but the open slot is fillable by ordinary NPs and the surrounding syntax is regular. *Memory* in *jog memory* is interpretable as the same noun that appears elsewhere; *jog* contributes a metaphor of brief shaking. The construction-specific meaning ("cause to recall briefly") partly overrides the compositional reading, but the parts remain visible. Schema-wise: the construction is the mediator; the parts' meanings remain partly accessible; the relation is interpretation, but the interpretation is partly construction-specific.

Frozen idioms (*kick the bucket*) test the lower bound. The parts contribute nothing to the construction's meaning of "die"; *kick* and *bucket* are dead lexically inside the construction even if they remain alive elsewhere. At this end of the gradient, P (the parts' meanings) is no longer accessible through X (the construction) for R (interpretation). The construction passes the schema's mediator-of-form requirement (it has internal structure that *could* have mediated access) but fails the accessibility-remains-through-it test. Frozen idioms mark where transparency in the umbrella sense bottoms out for the interpretation relation.

The gradient isn't unique to argument-structure or idiomatic constructions; it runs through morphology too. Productive prefixation (*un-* in *unhappy*) sits near the fully compositional end. Partly opaque derivatives (*business* from *busy*) sit near the idiomatic end: the historical compositional source is recoverable for analysts but not for ordinary speakers. The same gradient logic recurs in §4 (compound transparency) and §5 (priming-graded morphological transparency), though the domains and operationalizations differ.

3.3 Constructional transparency as a network property

Section 3.2 treated each point on the gradient as a property of an individual construction. The present paper develops a stronger claim that builds on Goldberg's framework without being orthodoxy in it: constructional transparency also depends on the network in which a construction sits. Inheritance hierarchies link broad generalizations to subregularities to exceptional low-level constructions (Goldberg, 2006, p. 14). A construction's parts and their compositional contribution are accessible partly because related constructions in the network make them so.

Goldberg's worked example: the *What's doing ?* construction (*What's that fly doing in my soup?!*) has a fixed form and connotes unexpectedness; it inherits from the Left Isolation construction, the Subject-Auxiliary Inversion construction, the Subject-Predicate construction, and the Verb-Phrase construction (Goldberg, 2006, p. 14). A learner who treats *What's the* X *doing* Y*?!* as a construction in its own right does so partly because each of its inherited constructions has its own attestations and contributes a piece of the form–meaning organization. Without the inherited constructions in the network, the learner would have less to work with.

This makes constructional transparency a network property in two distinct senses. Interpretive transparency: a construction's compositional reading is accessible partly because related constructions provide compositional support. The Ditransitive's transfer meaning is interpretable partly because the *for*-dative paraphrase (*Liza bought a book for Zach*) and the regular argument-structure constructions provide compositional context (Goldberg, 2006, p. 10). Learnability transparency: a construction is acquirable partly because its place in the inheritance network gives the learner multiple routes of access. Both senses are R-relativized in the umbrella schema: P remains accessible for the purposes of R, where R is interpretation in the first sense and learning or analogical generalization in the second.

The network claim is offered as a hypothesis, not as established. If the hypothesis holds, constructions in dense regions of the constructicon would be more transparent for both R-values than constructions in sparse regions, and constructions whose family-resemblance relatives have low type frequency would be harder to interpret and learn. Goldberg (2006, p. 15) cites usage-based acquisition work (Diessel and Tomasello, Tomasello, Saffran and colleagues) as part of the broader case for usage-based acquisition; whether any of those studies bear specifically on transparency-as-network-property is itself an empirical question the present paper leaves open. The form–meaning accessibility family of §§3–5 will return to the hypothesis: the corpus-derivable predictors that Bell and Schäfer use in §4 and the priming-magnitude modulations that Heyer and Kornishova report in §5 will be treated as compatible quantitative probes for the network-relative transparency that §3 frames theoretically, not as direct measurements of it.

The network claim also clarifies what frozen idioms mark. *Kick the bucket* has nearly no parts-to-whole compositional accessibility, but speakers recognize it as an idiom rather than as random word combinations. That recognition is partly supported by the existence of partly-idiomatic constructions one rung up the gradient: network position helps explain why the string is treated as a unit. The construction can also be stored holistically and accessed without compositional decomposition; nothing in the network claim rules that out. Even at the bottom of the gradient, network position does some work, but the work is recognition-relative, not interpretation-relative.

3.4 Number-transparent NPs revisited

Section 2 treated CGEL number-transparent NPs as agreement-accessibility cases. From the CxG side, those same NPs are constructions: form–meaning pairings whose form is the number-transparent quantificational noun + *of*-complement template and whose meaning includes the agreement override. CGEL preserves head-and-complement structure (with *lot, number*, etc. as heads) and treats the override as a property of the construction rather than a separately stipulated mechanism (the term *percolation* appears only informally in a footnote (Huddleston & Pullum, 2002, §18.2 fn. 12)). CxG variants differ on whether head-and-complement structure is preserved internally: SBCG preserves it explicitly; some Goldbergian variants leave the internal representation more abstract. The choice is methodological; the umbrella claim survives under any of them.

This connects §2 and §3 in a useful way. Number-transparent NPs are constructional-transparency cases, but the relation R is agreement (a §2-style morphosyntactic relation), not interpretation (the §3-style relation). The umbrella schema is one frame; the senses partition by R. A construction can be transparent for one R and opaque for another: a frozen idiom like *kick the bucket* is opaque for interpretation but morphosyntactically regular (the verb agrees with its subject in the ordinary way). R-relativity is built into the umbrella; it's a feature, not a bug. The schema asks whether P remains accessible *for the purposes of* R, and accessibility isn't transitive across R-values.

The methodological consequence: choosing CGEL's apparatus or CxG's apparatus for any given case is a question about which formal vocabulary makes the empirical generalizations clean. The deeper question of which framework is "right" isn't at issue here. The umbrella schema sits above both: it states the relational claim that a transparency analysis has to satisfy, without committing to how the satisfaction is formally encoded.

A coherence point across the frameworks the paper engages: CGEL, SBCG, and Goldbergian CxG are all non-derivational, and none posits empty mediating elements (traces, copies) between a surface position and a hypothesized underlying one. Contemporary dependency-grammar accounts (da Cunha & Gibson, 2025; Gibson, 2025) take the same line on long-distance dependencies, arguing that direct head-to-dependent connections fit acceptability and processing data without empty mediators (cf. the §2 extraction-transparency footnote). The umbrella schema doesn't adjudicate the dispute, but the frameworks engaged here all treat the mediator X as observable structure rather than as posited invisible mediation between observable layers. Where the schema asks whether P remains accessible through X for R, the candidate X is always something the analyst can point at.

3.5 Transparent free relatives revisited

The TFR agreement/syntactic-category dissociation is constructional transparency in a tight sense: a constructional sub-type (the attributional daughter) has access conditions that the parent lacks, and the access condition tracks an inheritance relation rather than a per-item lexical property. The TFR-parent licenses agreement transparency across the full verb inventory; the TFR-attributional daughter licenses syntactic-category transparency in addition. Kim's (2026, ch. 6.8) treatment is the architectural anchor; the corpus check in Kim and Reynolds (2026) supplies the parent/daughter split's empirical content. The two competing readings of the daughter's licence (§2.5: metalinguistic-attribution vs. mention-mediated) are both constructional in §3's sense; the work §3.5 does is independent of which reading wins.

The TFR case gives §3.3's network hypothesis a tractable instantiation, though not a confirmation. The daughter's predictions (verb-class restriction, PP-restriction to characterizing PPs, the *take to be* placement that defeats a copula-as-blocker analysis) are stabilized by the construction's position in an inheritance network, rather than by any verb's lexical entry or by the matrix's selectional requirements. R-relativity is also visible: the same mediator is transparent for agreement (both daughters) and only partially transparent for syntactic category (daughter-only). The CGEL-vs-CxG choice flagged in §3.4 reappears: an inheritance-hierarchy framework like SBCG (Kim's choice) makes the parent/daughter relation the central analytical move; a Goldbergian variant absorbs the same relation into a network of family-resemblance constructions.

3.6 Looking ahead to §§4–5

Section 3 has set up the theoretical frame for the form–meaning accessibility family: constructions as form–meaning pairings, compositionality as gradient, transparency as network property. Section 4 will give the quantitative measurement. Bell and Schäfer's (2016) models of compound transparency operationalize expectedness via corpus statistics and human ratings, providing empirical content for the partly-schematic and idiomatic regions of the gradient. Section 5 will give the processing measurement. Heyer and Kornishova (2018) show that semantic transparency modulates masked priming, but only at long stimulus-onset asynchronies, a finding that bears on the time-course of network access.

The three sections together motivate the form–meaning accessibility family, with three evidential windows that bear on it; whether they converge into a cross-domain projectibility cluster is what §7 will weigh, and §7 doesn't claim convergence delivered. Whether each within-domain profile passes Slater's (2015) SPC test, and whether Khalidi's (2018) causal-network refinement applies to any cross-domain reading, are questions §7 returns to. For now, §3 has done its job by setting the frame: constructional transparency is gradient and network-relative, and the next two sections give the family its quantitative and processing-experimental bone structure.

4 Semantic transparency

Section 3 set up form–meaning transparency as a theoretical gradient and a network property. Section 4 narrows R to interpretation and turns to a quantitative case: rated transparency of English noun-noun compounds. Section 4.3 places referential transparency as the schema's boundary case, where X is a context rather than a form.

4.1 Compound transparency: the Bell and Schäfer model

Bell and Schäfer (2016) model the perceived semantic transparency of English noun-noun compounds as a function of *expectedness* in their internal semantic structure. A compound is rated as more transparent when its semantic structure is more expected, both in the predictability of its constituents in their positions and in the predictability of the semantic relation between the constituents given those constituents. Transparency is treated as a scalar variable rather than a binary or four-fold property. The empirical anchor is the database of Reddy and colleagues (2011): 30 ratings on a 0–5 scale for each of 90 two-part NN compounds, totalling 8,100 ratings (Appendix B.5).

Bell and Schäfer's models identify three predictors of higher rated transparency. First, N1 *frequency*: compounds whose first noun occurs more frequently in the language overall are rated as more transparent. Second, *semantic-relation expectedness given* N1: compounds whose semantic relation is more frequently associated with that particular N1 are rated as more transparent. Third, N2 *productivity*: compounds whose second noun occurs more often as the head of NN compounds are rated as more transparent. The first and third predictors are corpus-statistical; the second requires hand-coding of semantic relations using Levi's (1978) nine-predicate inventory (CAUSE, HAVE, MAKE, USE, BE, IN, FOR, FROM, ABOUT; Appendix B.1) and aggregation across compounds sharing each N1.

The two headline conclusions: N1 and N2 contribute asymmetrically to whole-compound transparency, and all significant predictors of rated transparency also independently predict processing speed. The latter motivates Bell and Schäfer's tentative hypothesis that "perceived transparency may itself be a reflex of ease of processing" (Bell & Schäfer, 2016, p. 157). The hypothesis is offered as a direction for further work, not a result. Stress-pattern speculation about the IN/FOR positive associations is in Appendix B.3; the N1/N2 asymmetry detail is in Appendix B.4.

4.2 Compound transparency in the umbrella schema

The umbrella schema's variables map directly. X (the mediator) is the compound, e.g. *car-wash*, *jailbird*, *hogwash*. P (what remains accessible) is the constituent meanings and their semantic relation. R (the relation) is interpretation: rated transparency operationalizes how successfully the constituents' meanings reach interpretation through the compound. The schema's "remains accessible" requirement is gradient here, just as in §3: a compound's parts are accessible to varying degrees, with rated transparency as the empirical proxy.

Bell and Schäfer's three predictors fit naturally inside the schema. N1 frequency, N2 productivity, and semantic-relation expectedness given N1 are properties of the network in which a compound sits, not properties of the compound in isolation. A compound is rated more transparent when its constituents and the relation between them are more conventional in the language. This is the network-relative view §3.3 framed theoretically, expressed quantitatively. Compound transparency is a property of the compound's position in the constructicon-and-corpus, not of the compound alone.

Bell and Schäfer's predictors are corpus-statistical; the constructicon network is a theoretical posit. Frequency-driven entrenchment is the standard usage-based story for how the constructicon is shaped, but Bell and Schäfer don't directly measure constructicon-network properties. Their work is consistent with §3's network claim without testing it.

4.3 Referential transparency as boundary case

The schema's referential boundary is independent of the form–meaning accessibility family. *Lois Lane believes Superman is brave* is a textbook intensional context; substituting *Clark Kent* for *Superman* yields a sentence whose truth-conditions may not be preserved (the de re reading allows substitution; the de dicto reading doesn't). This is referential transparency in the philosophical-semantics tradition, and the schema technically applies: X is the intensional context (the embedding under *believe*), P is the referent of the embedded singular term, R is substitution under co-reference.

What makes this the boundary case is that X is a *context* rather than a *linguistic form*. In all the cases §§2–3 worked through, X was a form whose internal structure could have mediated access to P: a quantificational-noun-of-complement construction; a compound; a partly-schematic argument-structure construction. An intensional context isn't a form in that sense. It's a configuration that creates substitution-opacity, and the de re/de dicto distinction is about whether substitution is allowed, not about whether form-internal compositional access is preserved.

The schema therefore stretches thinnest at this boundary. P is a referent, not a constituent meaning or a number feature; R is substitution-salva-veritate, not agreement or interpretation. The umbrella applies on a sufficiently abstract reading: referential transparency is a kind of mediated accessibility, where the mediator is the intensional embedding and the referent is what does or doesn't remain accessible. But the difference in the type of X is real and matters for the metaphysical conclusion. Referential transparency isn't part of the form–meaning accessibility family of §§3–5.

The philosophical literature on referential transparency (Quine (1956) as canonical anchor; the Frege/Kripke/Kaplan tradition) would dominate the section if developed. The boundary marks a real heterogeneity in the umbrella, not a defect: the schema's R-relativity handles the case without forcing it into the form–meaning family.

4.4 Compound transparency as a within-domain projectibility profile

Compound transparency joins constructional compositionality (§3) as a within-domain projectibility profile on form–meaning accessibility. Class-membership talk doesn't apply at this level: there's no small, conventionally stable class of "transparent compounds" comparable to number-transparent quantificational nouns. The story here is gradient: compounds are more or less transparent on a continuous scale, and the predictors of higher transparency are corpus-statistical and constructional.

Compound transparency licenses non-trivial within-domain projection. Knowing a compound's N1 frequency, semantic-relation expectedness, and N2 productivity lets us predict its rated transparency, with non-trivial fit in Bell and Schäfer's models. The stabilizing source is usage-based entrenchment: the network of compounds and their semantic relations is shaped by frequency, productivity, and the kind of conventional patterns that emerge from repeated use. Falsification condition: if N1 frequency, semantic-relation expectedness, and N2 productivity failed to predict rated transparency in an extension or replication of Bell and Schäfer's models, the within-domain profile would fail. Whether the within-domain projection joins with §3's theoretical profile and §5's processing window into a cross-domain projectibility cluster on shared items is left to §7, which doesn't take that result as delivered.

What §4 contributes beyond §3 is empirical content. The compositionality gradient theorized in §3 is a measurable property whose predictors are corpus-derivable and whose values correlate with rated transparency. That correlation is what makes compound transparency a quantitative window onto the gradient. Whether the same predictors would also track processing latencies, which would strengthen the case for Khalidi-style causal-hierarchical refinement (without by itself establishing it), is a question for §5.

4.5 Bridge to §5

Section 5 takes the third window on the family: time-sensitive lexical processing. Where Bell and Schäfer measure rated transparency, Heyer and Kornishova measure masked priming. Where rated transparency is a steady-state property, priming is a time-course property. Whether the windows of §3, §4, and §5 jointly support the kind of causal-hierarchical projection that Khalidi's framework formalizes is the question §7 will return to.

5 Processing transparency

Section 4's rated transparency is a steady-state property of the compound. Section 5 turns to a time-course property: how does semantic transparency modulate the processing of morphologically complex words during recognition? Heyer and Kornishova's (2018) masked-priming study is the empirical anchor. Their finding, summarized in their title's three-dot ellipsis: semantic transparency affects morphological priming, but only at long stimulus-onset asynchronies.

5.1 The Heyer and Kornishova study

Heyer and Kornishova (2018) ran two parallel masked-priming experiments, one with English *-ness* nominalizations and one with Russian *-ost'* nominalizations. The two suffixes are deadjectival and produce abstract nouns denoting "state of being X" (e.g., *dark* + *ness* = 'darkness'; *bledn-* + *-ost'* = *blednost'* 'paleness'). Targets in both experiments were the adjective bases (e.g., *pale, dark*); primes were the corresponding nominalization, a simplex unrelated control, or the adjective as identity prime.

Each experiment crossed two stimulus-onset asynchronies (SOAs), short and long, with transparency as a continuous variable rated by separate groups of speakers (Appendix C).

Transparency was rated on a 1–7 scale: *business* (rated 3.17) at the opaque end is no longer interpretable as "state of being busy"; *falseness* (rated 5.75) is fully derivable from *false*; items spanned the range continuously.

The headline result is SOA-conditioned. At short SOA in both languages, priming is constant across the transparency scale: morphologically complex primes are decomposed into stems regardless of whether the stem still contributes to the meaning of the whole. At long SOA, priming becomes transparency-modulated: more transparent prime-target pairs (e.g., *paleness-pale*) yield larger priming effects than less transparent pairs (e.g., *business-busy*). The three-way interaction (Prime Type × Transparency × SOA) is significant in English and marginally so in Russian, with the same direction in both languages (Heyer & Kornishova, 2018, pp. 1117–1119). Statistical detail is in Appendix C.3.

5.2 The serial-processing interpretation

Heyer and Kornishova read the data as evidence for a serial decomposition process. Morphologically complex words are first decomposed into stem-plus-affix on the basis of orthographic structure (*morpho-orthographic decomposition*), with semantic information entering only at a later stage. The first stage is semantically blind: it parses *paleness* into *pale* + *ness* regardless of whether the parse yields a coherent semantic interpretation, and would parse *corner* into *corn* + *er* even though the parse fails semantically (Heyer & Kornishova, 2018, p. 1112).

The pattern is consistent with Bell and Schäfer's (2016) hypothesis that "perceived transparency may itself be a reflex of ease of processing." Within the masked-priming paradigm, fast-and-shallow windows show no transparency modulation; longer windows do. The two studies probe different items (NN compounds, derivational nominalizations) and different methods (ratings, masked priming), so they don't test the same processing chain; they're compatible probes for a shared hypothesis.

5.3 The rival parallel-processing position

The serial reading is contested. Feldman and colleagues' parallel-immediate position argues that morpho-semantic information influences recognition from the earliest stages. Feldman et al.'s (2015) time-course analysis presses the diagnostic question (*must analysis of meaning follow analysis of form?*) with a same-target design (each target is compared after semantically similar and semantically dissimilar primes within subjects) and SOA modelled as continuous from 34 to 100 ms rather than as binned categorical levels. They report that semantic similarity affects recognition latencies even at SOAs of 34 ms and 48 ms, shorter than Heyer and Kornishova's *short* windows of 39 ms (English) and 33 ms (Russian). On the parallel-immediate reading, the form-then-meaning ordering doesn't survive that finding. The two designs aren't head-to-head: Heyer and Kornishova's short-SOA result is the absence of transparency modulation across an item set rather than the absence of semantic effects on a same-target comparison, and the disagreement is whether semantic information is constitutively present at masked-priming windows or emerges only after morpho-orthographic decomposition. The literature is mixed enough (the meta-table on p. 1121 of Heyer and Kornishova surveys 35 prior masked-priming studies) that the present paper takes the serial reading as Heyer and Kornishova's reading, not as a settled disciplinary verdict. The cross-modal probe in Appendix D would generate within-items evidence relevant to the disagreement.

5.4 What §5 contributes to the cluster

Section 5 contributes the time-course dimension to the form–meaning accessibility family. Section 3 framed compositional transparency as a gradient; Section 4 made it quantitative through corpus predictors and ratings; Section 5 makes it temporally articulated. The family's evidential support is now: a theoretical scale, a steady-state quantitative measure, and a time-course measure. The three windows are operationally distinct but conceptually related: they're different probes for the same underlying form–meaning accessibility, taken at different stages of speakers' interaction with the relevant items. Falsification condition for the within-domain priming profile: if rated semantic transparency failed to modulate masked-priming magnitude at long SOAs in an extension or replication of Heyer and Kornishova's design, the profile would fail.

Bell and Schäfer's compounds and Heyer and Kornishova's nominalizations are different items in different morphological domains; their results don't compose into a transitive empirical chain on the same items. They're compatible probes for the form–meaning accessibility hypothesis, not direct measurements of one another. Whether the cluster hypothesis survives the within-items test (corpus predictors, ratings, priming magnitude, and interpretation accuracy aligning on shared items as joint reflexes of underlying representational structure) is the empirical question Appendix D sketches as a probe design. Until that probe is run, the family is three within-domain profiles plus a research hypothesis, not a delivered cross-domain cluster.

5.5 Cluster status and bridge to §6

Sections 3–5 give three within-domain projectibility profiles stabilized by different sources: constructional inheritance (§3), usage-based entrenchment (§4), processing architecture (§5). Whether they compose into a cross-domain cluster on shared items is for §7, with the probe design in Appendix D. Acquisition trajectory is a plausible further extension consistent with §3.3's network claim but not directly grounded by the present sections. Section 6 turns from form–meaning accessibility to the comparative question of §1.

6 Typological transparency

Section 6 is methodological. The previous sections worked through cases in English and Russian; Section 2.6 names Korean case-stacking and scrambling as a deferred extension rather than as evidence in the present paper. How should the paper handle the move from those language-particular cases to any cross-linguistic claim? Haspelmath's (2010) descriptive-versus-comparative-concept distinction is the right discipline.

6.1 Haspelmath's distinction

Haspelmath (2010) argues that crosslinguistic comparison shouldn't assume universally available descriptive categories like ADJECTIVE, PASSIVE, ACCUSATIVE, or SUBJECT. Each language has its own descriptive categories, defined by the language-particular distributional and morphosyntactic criteria that make the categories work in that language (Haspelmath, 2010, p. 664). The criteria differ from language to language: what counts as a passive in one language needn't be what counts as a passive in another, so descriptive categories can't be equated across languages.

Comparative concepts, by contrast, are analyst-created tools designed for the purpose of cross-linguistic comparison. They aren't part of any language system and aren't psychologically real; they can only be more or less well suited to the comparative task, rather than right or wrong (Haspelmath, 2010, p. 665). Comparative concepts have to be universally applicable, which means they have to be built from universally applicable building blocks: conceptual-semantic concepts (such as notions of agency or possession), general formal concepts (such as "phonological word"), and other comparative concepts.

Haspelmath calls the alternative position categorial universalism: the assumption that there's a substantial set of crosslinguistic descriptive categories from which languages select (Haspelmath, 2010, p. 663). Newmeyer's (2007) *Linguistic typology requires crosslinguistic formal categories* is the canonical statement of the position Haspelmath rejects: typological generalizations, on Newmeyer's argument, can't be made through purely conceptual-semantic comparative concepts because such generalizations depend on reference to the specific form in which the concepts are realized (Newmeyer, 2007, p. 136). Haspelmath (2010, p. 666) grants the realization point but argues that the relevant formal references are themselves comparative concepts (e.g. "adposition", "zero coding") definable in conceptual-semantic and general-formal terms, not crosslinguistic descriptive categories. Categorial particularism, his preferred position, holds that descriptive categories are language-particular and that comparison happens through analyst-defined comparative concepts that stand in many-to-many relations with descriptive categories. The position has been developed in typological work by Dryer, Croft, Lazard, and Cristofaro.

The methodological consequence is straightforward. A typological generalization isn't a claim about a descriptive category instantiated in every language. It's a claim about how analyst-defined comparative concepts are realized in language-particular ways. The generalization that nominative case-marking is widespread across languages doesn't entail that "nominative" picks out the same descriptive category in any two languages.

6.2 The umbrella schema as a comparative concept

The umbrella schema introduced in §1 is a comparative concept. It's built from conceptual-semantic notions (mediation, accessibility, property, relation) that are universally applicable, plus relational variables (X, P, R) whose values are filled in language-particularly. The schema's question (X *is transparent w.r.t.* P *for* R *when* P *remains accessible through* X *for the purposes of* R) is something the analyst asks of any language; it's not a claim that any particular language has a descriptive category corresponding to TRANSPARENCY.

The schema's primitives meet Haspelmath's (2010, p. 665) standard for comparative concepts (conceptual-semantic notions, general formal notions, other comparative concepts). *Property* (P) and *relation* (R) are conceptual-semantic primitives. *Form* (X) is a general formal notion. *Accessible through* is the contested slot. Accessibility has discipline-specific senses (discourse-pragmatic salience, psycholinguistic retrievability, formal-semantic substitution-licensing); the schema's sense is more abstract: P is accessible to R through X when R tracks P across X rather than tracking some property of X itself. The notion is conceptual-semantic, not psycholinguistic. The case studies operationalize it differently (agreement targeting, constructional interpretability, rated transparency, priming magnitude), but the comparative concept lives at the abstract level. *Mediation* is the schema's other central notion, tested by the contrast between paradigm cases and the §2.3 non-instance.

Each value of the schema's variables is language-particular. In English, X is the number-transparent quantificational noun + *of*-complement construction (a CGEL descriptive category); P is the number/count profile of the complement; R is subject-verb agreement. In the Korean extension named in Section 2.6, X may be a layered case-marking pattern or a scrambled word order; P may be argument-role assignment, scope, or information-structural status; R may be argument-role recovery, scope interpretation, or discourse licensing. The variable bindings are different. The comparative concept MEDIATED ACCESSIBILITY is the same.

Haspelmath's framework requires exactly this: cross-linguistic claims are made through comparative concepts whose realizations differ from language to language. The English and Korean cases aren't "instances of the same descriptive category." They're different descriptive categories that can be compared through the comparative concept of mediated accessibility, with different X, P, and R values.

The form–meaning accessibility family of §§3–5 isn't itself a comparative concept. The schema (mediated accessibility, with its $X/P/R$ variables) is the comparative concept; the family is a set of language-particular phenomena and theoretical/experimental measurements compared *through* the schema. CxG constructional transparency, compound rated transparency, and morphological priming are different operational windows on form–meaning accessibility (§§3–5 worked them through in English with one Russian extension): a theoretical profile, a measurement, and an experimental effect. They're intended to bear on the same comparative-concept gradient. The cross-linguistic move at the comparative-concept level doesn't commit the paper to crosslinguistic descriptive categories.

6.3 Worked example: descriptive category and comparative concept

Consider the typological question: do all languages have number-transparent constructions? The categorial-universalist answer is to look for English-style number-transparent NPs in every language. The categorial-particularist answer is different. NUMBER-TRANSPARENT QUANTIFICATIONAL NOUNS is a CGEL English descriptive category, defined by the override behaviour (near-obligatory in the central cases), the small and relatively stable membership, the characteristic determiner patterns, and the partitive/non-partitive *of*-complement restrictions worked through in §2.1. Crosslinguistic comparison tracks the comparative concept *organizing* that descriptive category, not the descriptive category itself.

What the comparative concept of mediated accessibility lets us ask is different. Across languages, which configurations let an internal property remain accessible to a grammatical relation that the surface form might otherwise have hidden? English number-transparent NPs answer the question for the relation R = subject-verb agreement, with P = number/count profile of the *of*-complement. The comparative concept doesn't predict that any other language has matching descriptive categories. It predicts that the analyst's question is askable of any language, and that the answers will differ.

The cross-linguistic extensions the paper has on hand are limited. Section 5's English *-ness* / Russian *-ost'* parallel is one: two languages, two descriptive categories (deadjectival nominalization in English; deadjectival nominalization in Russian), but the same comparative-concept question asked of both, with parallel answers. The Korean extension in Section 2.6 is the most consequential next test, deferred for separate co-authored development. Without those Korean answers, this section's worked example carries the methodological move with English alone; the cross-linguistic claim is a structural one (the comparative concept lives above the descriptive category, whatever the language) rather than a fully demonstrated comparative one.

The claim sits at the right crosslinguistic level. It requires neither a universal descriptive category TRANSPARENCY or NUMBER-TRANSPARENT nor a claim that any two languages have "instances of the same construction." It claims, more modestly, that the comparative concept of mediated accessibility can be applied to any language and that the answers it elicits are language-particular categorial content. That modesty is what categorial particularism asks of comparative work.

6.4 The reviewer objection and a reply

A reviewer might object that the umbrella schema looks like a universal descriptive category smuggled in as a comparative concept. The paper holds that TRANSPARENCY is a family of senses rather than a single kind, but the umbrella schema is intended to apply across English, Korean, and the form–meaning family of §§3–5; is that not exactly what categorial universalism claims?

The reply is that the schema is a comparative concept by design. It's an analyst's question (does P remain accessible through X for R in this language?), not a claim that any language has a descriptive category called TRANSPARENCY. Each language answers the question differently or not at all. The schema's universal applicability is the universal applicability of the *question*, not of any *answer*. Haspelmath's (2010, p. 665) criterion for a good comparative concept (it can be more or less well suited to the task of permitting crosslinguistic comparison) is exactly the standard the umbrella schema is meant to meet.

Newmeyer's (2007) specific argument deserves a direct response. Newmeyer holds that typological generalizations have to reference language-particular formal realizations, not just conceptual-semantic concepts, and that this commits typology to crosslinguistic formal categories. The umbrella schema's variables address this without conceding the categorial-universalist conclusion. The schema's X slot is filled language-particularly: the English X is the number-transparent quantificational noun + *of*-complement construction (a CGEL English descriptive category); the English TFR X is the fused-relative construction with attributional-verb licensing (a CGEL/CxG English descriptive category); the Korean X (Section 2.6) is whatever Korean construction the analyst identifies. The schema's universal applicability is at the analyst-question level; the formal realizations Newmeyer demands enter at the variable-binding level, language-particularly, without becoming crosslinguistic descriptive categories. The position is closer to Haspelmath's (2010, p. 666) reply to Newmeyer (formal references can be comparative concepts in their own right) than to the categorial-universalist alternative.

The metaphysical conclusion in §7, that the umbrella schema is a tool for locating projectibility profiles rather than itself a kind, is consistent with this. Three levels stay distinct: the comparative concept (mediated accessibility) lives at the cross-linguistic methodological level; descriptive categories (English number-transparent quantificational nouns, Korean case-stacking and scrambling, and so on) live at the language-particular grammatical level; PROJECTIBILITY PROFILES are observable bases licensing non-trivial projection, stabilized by identifiable sources, realized within or across such descriptive categories. Kindhood is a downstream verdict on profiles that earn it. The paper doesn't collapse the levels.

6.5 Methodological default

Haspelmath's distinction has been contested in the typological literature; the present paper doesn't adjudicate. Haspelmath's position is taken as the safer methodological default for cross-linguistic claims, since it requires no commitment to descriptive-category universality and doesn't assume a substantive crosslinguistic ontology of grammatical categories. Readers more comfortable with categorial universalism can read the umbrella schema as a useful working concept whose status is secondary to its methodological pay-off.

Croft's Radical Construction Grammar (RCG) is the construction-grammar variant closest to the present categorial-particularist commitment. RCG routes cross-linguistic comparison through language-particular constructions linked by similarity rather than through universal grammatical categories or formal apparatus. The umbrella schema's relation between comparative concept and language-particular descriptive category is broadly compatible with RCG's architecture: language-particular constructions instantiate the schema's variables (X, P, R) without committing the analyst to a universal descriptive category of transparency. The choice between routing through Haspelmath-style comparative concepts and routing through RCG's construction-similarity hierarchies is left as an option for further development; both preserve the categorial-particularist commitment.

6.6 Bridge to §7

Section 7 develops the metaphysical claim: the umbrella schema locates projectibility profiles, each stabilized by an identifiable source (lexical convention, constructional inheritance, usage-based entrenchment, processing architecture, formal-semantic environment); Slater's (2015) SPC framework and Khalidi's (2018) causal-network account are diagnostic tools for assessing projection-supporting stability rather than master ontologies. The claim is consistent with categorial particularism: descriptive categories are language-particular; the comparative concept MEDIATED ACCESSIBILITY lives at the cross-linguistic level; PROJECTIBILITY PROFILES are observable bases for projection, realized within or across descriptive categories. Cross-linguistic projection at the profile level isn't entailed by cross-linguistic projection at the comparative-concept level.

7 Projectibility profiles

Section 1 set the paper's central question: when does a transparency attribution support induction, and what stabilizes that induction? In Boydian terms, the question is field-relative: a profile is projectible for a field's explanatory and inductive tasks (Boyd, 1991, 1999). The primary field here is syntax. The strongest profiles support induction over agreement, syntactic-category licensing, complement selection, and related grammatical behaviour.

Section 7 collects the answers and keeps the field-relative claims separate. The umbrella schema is a tool for locating projectibility profiles, not itself a profile; the profiles the paper has examined are stabilized by heterogeneous structures; Slater's (2015) SPC framework and Khalidi's (2018) causal-network account are complementary diagnostics rather than master ontologies; kindhood is a downstream verdict.

7.1 The umbrella isn't a single profile

Section 6 established that the umbrella schema is a comparative concept, an analyst's tool for asking the projectibility question of any language. The metaphysical question is what *follows* from posing the question across a heterogeneous family of cases. The umbrella's heterogeneity is the answer. The X-values across the family include lexical classes (number-transparent quantificational nouns), constructions (the ditransitive, partly-filled idioms), compounds (NN compounds), derivationally complex words (*-ness* and *-ost'* nominalizations), and intensional contexts (referential transparency). The P-values include number/count features, constituent meanings and their relations, compositional contributions, lexical-stem identity, and referents. The R-values include subject-verb agreement, interpretation, masked priming, and substitution under co-reference. No single stabilizing source unifies these triples, and no single field's inductive base supports projection across them.

The umbrella identifies many candidate profiles, not one. The schema doesn't by itself license projection: knowing that a case instantiates the schema doesn't let us predict anything substantive about it, because the projectible content lives at the level of the individual profile, not the level of the schema. The schema's role is to make the projectibility question askable, case by case.

A useful general distinction follows from this heterogeneity. A *distributional class* names a real grouping at the surface (the schema's $X/P/R$ slots get filled by some real configuration), while a *projectibility profile* additionally requires some identifiable mechanism keeping the cluster projectible. The umbrella schema names a distributional class but not a projectibility profile: the schema picks out a real family of cases, but no mechanism unifies them at the umbrella level. Each profile examined below has to be evaluated separately for whether it qualifies as projectible in this stronger sense. The negative-polarity-item literature offers a parallel pattern: NPIs share a surface licensing property but the licensing patterns Hoeksema catalogues aren't manifestations of one mechanism, so NPIs are a class but not a projectibility profile at the umbrella level (developed at length in (Reynolds, 2026b, ch. 7)).

7.2 Syntactic and adjacent profiles

The paper's primary profiles are syntactic.

Number-transparent quantificational nouns (§2.4). The observable base is class membership, identified by lexical identity, quantificational semantics, corpus distribution as head of an *of*-complement construction, and typical determiner frame. The projected targets are non-trivial: near-obligatory agreement override in the central cases (in either direction); characteristic determiner patterns by subgroup; partitive vs. non-partitive *of*-complement availability; behaviour under fused-head construction; count/non-count percolation of the *of*-complement; restricted premodification. Class membership generates expectations for *new* members (boundary cases like *majority* and *minority*; gradient cases like *bunch*) that aren't built into the class's defining criteria. This is the strongest candidate profile examined here: the cluster supports inductive extension, not just cataloguing of the enumerated members.

Transparent free relatives (§§2.5, 3.5). The observable base is constructional sub-type and verb-class membership. The TFR parent licenses agreement transparency across the full verb inventory; the attributional daughter licenses syntactic-category transparency as well. Object-oriented attributional verbs (*call, consider, describe as, regard as, take to be*) predict AP-nucleus productivity and the narrower syntactic-category-transparency effect; the *seem*/*appear* class predicts agreement transparency without syntactic-category transparency. The corpus check is still working evidence, but the profile is syntactic in the same field-relative sense as number-transparent QNs: a constructional input projects to grammatical behaviour.

The adjacent form–meaning profiles belong to neighbouring fields rather than to syntax itself. The form–meaning accessibility family (§§3–5) contains three within-domain profiles, related by analytical theme rather than by demonstrated cross-domain projection. First, a construction-level profile: graded compositionality predicts compositional interpretability via inheritance and network position. Second, a compound-level profile: corpus-statistical predictors predict rated transparency. Third, a derivationally-complex-word-level profile: rated transparency predicts long-SOA priming magnitude. The construction-level profile is theoretical; the other two rest on direct empirical evidence (Bell & Schäfer's models; Heyer & Kornishova's experiments). Whether the three compose into a cross-domain cluster on shared items is an open empirical question; on present evidence they don't, and Appendix D sketches the within-items probe (corpus predictors, ratings, priming, interpretation accuracy on the same items) that would settle it. The form–meaning family is the open research programme of the paper, not a delivered result.

Acquisition trajectory is a plausible further extension consistent with the network claim of §3.3 but not directly grounded by the present sections; sourcing usage-based acquisition evidence is left for further work.

7.3 Stabilizing structures

The profiles are stabilized by different structures, and the paper's six R-values across §§2–5 (plus the boundary case of §4.3) call for a small inventory of stabilizing sources rather than a single mechanism.

Lexical convention stabilizes the number-transparent class as a class. Lexical inheritance here means that speakers acquire not isolated words but item-specific constructional slots: *a lot of* X, *plenty of* X, *the rest of* X, *a number of* X, with their associated agreement and complement-selection behaviour. The acquisition target is the slot, not just the lexeme. Speakers acquire a small, relatively stable inventory of these slots; new members enter at the boundary (*majority, minority*) or by sense-split (*couple*$_2$). Class identity itself isn't held by Boyd-style homeostatic feedback; it has the stability characteristic of any small, conventionally established grammatical class. The count/non-count percolation behaviour the class licenses connects to broader patterns in English countability; (Reynolds, 2026a) develops a Khalidian causal-cluster analysis of those patterns with conventional individuation as a central node. The transparency paper draws on that analysis but doesn't depend on its stronger claims; the within-NP percolation can be observed and predicted independently.

Constructional inheritance stabilizes the TFR daughter profile (§2.5) and the network-relative transparency hypothesis of §3.3. In TFRs, the parent and attributional daughter support different access conditions. In the form–meaning family, a construction's compositional accessibility depends partly on the inheritance hierarchy it sits in; constructions whose inherited supercategories are well-attested support stronger compositional projection than orphaned constructions.

Usage-based entrenchment stabilizes the corpus-statistical predictors of compound transparency (§4). Bell and Schäfer's frequency, productivity, and semantic-relation expectedness predictors are corpus-derivable measurements of usage entrenchment; the rated-transparency target is a steady-state property whose stability is plausibly a reflex of that entrenchment.

Processing architecture stabilizes the time-course modulation of priming (§5). Heyer and Kornishova's SOA-dependent transparency effect is a processing-architecture observation: the time-course of morphological decomposition makes morpho-orthographic information available before morpho-semantic information, and the resulting priming pattern depends on representational properties that the processing architecture exposes only at certain stages.

Formal-semantic environment stabilizes the referential transparency cases (§4.3). Substitution-salva-veritate behaviour depends on the logical structure of intensional contexts. This is the boundary case for the schema: the mediator is a context, not a form, and the stabilizing source is the formal-semantic structure of the embedding.

The list isn't closed. Other linguistic profiles may be stabilized by combinations of these or by sources the paper doesn't invoke (information-structural pressures, learnability constraints, normative/pedagogical regularization). The point is that "what stabilizes this profile?" is the right empirical question, and the answer is heterogeneous.

7.4 SPC and Khalidi as diagnostics

SPC and Khalidi are complementary diagnostics rather than nested ones. Slater asks whether the cluster is stable and projection-supporting and is neutral on *what* stabilizes it. Khalidi asks whether the stabilizing structure is itself causal-hierarchical, with named candidate core properties whose causal action generates derivative properties. A profile can satisfy SPC without satisfying Khalidi, and (in principle) the reverse. The PROJECTIBILITY PROFILE is the unit of analysis; the two diagnostics ask different questions of it.

Slater (2015) gives the stability-plus-projection diagnostic. A profile satisfies SPC when its cluster of properties is stable enough to support reliable induction, regardless of *what* maintains the stability. Slater's framework has affirmative content beyond maintenance-neutrality: cluster stability has to be cliquish (members co-occur reliably under counterfactually relevant conditions), and sub-cluster-to-cluster probabilistic entailment is what supports reliable projection. Number-transparent quantificational nouns are the strongest candidate to satisfy SPC examined here: the cluster is stable, the sub-group covariance documented in §2.4 (determiner-pattern, partitive availability, override-direction co-vary by sub-group) is the cliquishness SPC requires, and the maintenance is at least conventional/lexical for class identity (§7.3). The within-domain audit beyond override direction is empirically open; the cliquishness claim is licensed by the §2.4 sub-group covariance but not yet by an end-to-end corpus or experimental study.

TFRs are a second syntactic candidate. The agreement/syntactic-category split and the verb-class restriction form a projected cluster stabilized by constructional inheritance, though the present evidence is a working corpus check rather than a completed profile audit. Each form–meaning window (§§3–5) satisfies SPC within its own field. Whether the family as a whole satisfies SPC at the cross-domain level depends on the empirical question §7.2 leaves open and §D sketches as a probe design.

Khalidi (2018) applies where the stabilizing structure is ordered and causal-hierarchical. The account requires named candidate core properties whose causal action generates derivative properties via identifiable pathways. The strongest sub-case in the paper isn't QN class identity itself but the count cluster the class participates in via *of*-complement percolation. Reynolds (2026a) analyzes that cluster as Khalidian, with CONVENTIONAL INDIVIDUATION (speakers' construal of referents as discrete units, established by usage-based convention) as the candidate core property. The derivative properties causally linked to individuation include the count grammar's LOOSE features (plural agreement, the *many/few* contrast, plural-only forms) and TIGHT features (singular determiner *a*(*n*), low-cardinal selection, count/non-count compatibility on determiners and predicates). Loose features can be satisfied by individuation alone; tight features require the precise atomic enumeration individuation supplies under stronger interpretation.

The number-transparent QN construction extends this chain at the construction level. The QN's *of*-complement determines the count/non-count profile of the whole NP (*a lot of money* non-count; *a lot of tourists* count), projecting the complement's individuation profile to the QN's external interface for agreement, partitivity, premodification, and predicate selection. Within-NP percolation is the causal-hierarchical move at the construction-internal level; the count cluster is the causal-hierarchical move at the lexical-grammatical level. Both have core/derivative structure in Khalidi's sense, and the construction's design isn't an analyst's add-on: it's a constructional pathway that routes the complement's individuation to the matrix relation, with predictable behavioural signatures (the cluster of §2.4) attached.

For QN class identity, Khalidi-style refinement isn't motivated on present evidence: class identity itself is conventionally stabilized without a separate ordered causal hierarchy. SPC is the diagnostic that fits class identity; Khalidi applies one level out, where the class plugs into the count cluster.

For the form–meaning family of §§3–5, Khalidi is acknowledged as a possible refinement *if* corpus predictors, rated transparency, priming magnitude, and interpretation accuracy align on shared items as joint reflexes of underlying representational structure. The four windows aren't a temporal chain (masked priming probes early lexical access before explicit interpretation, and ratings are a late metalinguistic judgement) but candidate joint consequences of a single underlying organization. The within-items evidence doesn't yet exist; Appendix D sketches a probe design. Until that probe is run, Khalidi remains a possible refinement for the form–meaning family rather than a delivered analysis.

Kindhood, on this picture, is the downstream verdict. A profile that scores well on the SPC diagnostic, and especially one that also scores well on the Khalidi diagnostic, has earned the kind label. A profile that scores modestly is a candidate. A profile that fails either diagnostic isn't a kind, even if it remains a useful descriptive category. The diagnostic work is *what stabilizes projection*; the kind label follows.

The apparatus matters in two places. SPC's maintenance-neutrality lets the paper put conventional, entrenched, architectural, and formal-semantic stabilizers on the same footing, where Boyd's homeostasis would force one mechanism (a poor fit for lexical convention). Khalidi's causal-hierarchy diagnostic gives the paper a vocabulary for asking whether sub-profiles have internal ordering: the form–meaning family's possible cross-modal cluster is the candidate case.

7.5 The resulting claim

Linguistic TRANSPARENCY isn't a kind, and the umbrella schema doesn't aspire to be one. The schema is a diagnostic tool for locating field-relative projectibility profiles: cases where an observable base licenses non-trivial projection to a target, stabilized by some identifiable source. Different transparency profiles are stabilized differently (lexical convention, constructional inheritance, usage entrenchment, processing architecture, formal-semantic environment), and each profile's projectibility is what makes the transparency attribution scientifically useful, not the profile's classification within any single ontology of kinds.

Three levels stay distinct, per §6:

- COMPARATIVE CONCEPT: mediated accessibility, the cross-linguistic methodological tool defined by the schema *X is transparent w.r.t. P for R when P remains accessible through X for the purposes of R.*
- DESCRIPTIVE CATEGORY: language-particular grammatical content. English number-transparent quantificational nouns are a CGEL descriptive category; English NN compounds and *-ness* nominalizations are descriptive categories defined by language-particular criteria; Korean case-stacking and scrambling are Korean-particular descriptive categories; and so on.
- PROJECTIBILITY PROFILE: an observable base that licenses non-trivial projection to a target, stabilized by some identifiable source, realized within or across language-particular descriptive categories. This is the projectible profile of §7.1: a maintained-and-projectible cluster, contrasted with a *class* (mere distributional grouping). Profiles that score well on the SPC and (where appropriate) Khalidi diagnostics earn the local-kind label as a downstream verdict.

The contribution here is to apply the projectibility-first move to a *relational* concept. TRANSPARENCY is the relation X-mediates-P-for-R, not an entity. Asking *when does this attribution support induction, and what stabilizes the induction?* works for relations as it does for the umbrella concepts *language* and *grammatical category* addressed in related work. The paper asks the naturalization question for transparency, but answers it locally: transparency as an umbrella isn't a kind, while particular transparency profiles may be stabilized well enough to support induction. Kindhood enters the picture as a verdict on profiles that earn it, not as the master question that organizes the analysis.

7.6 Connections and limits

Other multi-sense umbrella concepts (e.g. *language*, *grammatical category*) admit the same projectibility-first move: the umbrella isn't itself a kind but specific senses can be projectible profiles. Reynolds (2026b, ch. 7) develops the structural analogue for negative-polarity items at length.

The Korean (§2.6) extension is the most consequential gap. If mediated accessibility fails to yield productive cross-linguistic generalization when applied to Korean case-stacking and scrambling, the typological discipline of §6 admits the negative result without sacrificing the comparative-concept frame.

8 Conclusion

Linguistic transparency is a family of mediated accessibility relations, not a single syntactic property. The umbrella schema (X *is transparent with respect to* P *for* R *when* P *remains accessible through* X *for the purposes of* R) is a comparative concept: an analyst's question, applicable to any language, asking which configurations let an internal property remain available to a grammatical or interpretive relation. It doesn't claim that any language has a descriptive category called TRANSPARENCY.

The main field-relative result is syntactic. Number-transparent quantificational nouns are the strongest candidate examined: a small lexical class whose membership predicts a cluster of correlated behaviours (agreement override, determiner patterns, partitive availability, fused-head behaviour, count/non-count percolation, restricted premodification), stabilized by conventional lexical inheritance. Transparent free relatives supply a second syntactic profile, with constructional sub-type and verb-class membership predicting agreement and syntactic-category transparency.

The form–meaning accessibility family (constructional compositionality, rated compound transparency, masked-priming morphological transparency) gives three adjacent within-domain profiles whose cross-domain integration on shared items remains an open research programme; Appendix D sketches the probe that would settle it. Coordination is excluded as a non-instance; referential transparency is acknowledged as the schema's boundary case.

Korean case-stacking and scrambling (§2.6) are deferred for separate co-authored development. Acquisition trajectories are a plausible extension the paper doesn't ground.

A transparency attribution is scientifically useful when it identifies a field-relative projectibility profile: an observable base that licenses non-trivial projection to a target, stabilized by an identifiable source. Some profiles support strong projection; others weak; some none. Kindhood is the downstream verdict on profiles that earn it, not the master question of the analysis.

A Number-transparent NP empirical apparatus

This appendix collects the COCA pilot data behind the §2 claims about the number-transparent quantificational noun class. The body of the paper carries the qualitative claims; the audit trail lives here. Full machine-readable data: `paper/data/coca-pilot.md`; KWIC spot-checks: `paper/data/kwic-checks.md`.

A.1 Methodology

Queries were issued through the local Playwright wrapper at `tools/english-corpora/bin/ecorg.mjs` (May 2026). Each query is a surface string; counts are raw COCA frequencies, not filtered for head versus modifier use, idiom contamination, or sense disambiguation, except where noted. Two filtering issues recur:

- Modifier confound. Surface strings of the form X *of* N may include cases where N is a modifier inside a larger NP rather than the head of the *of*-complement (e.g., *lots of day-care* rather than *lots of day*). The countability paper (Reynolds, 2026a) handles the same coding issue for *three police* and *three cattle*. Where small counts fall in the predicted-counter direction, the modifier confound likely inflates them.
- Parse-shift. Surface strings containing a number-transparent QN plus a verb form may include cases where the QN sits inside a relative clause modifying a different head-noun, with the visible verb agreeing with that other head. *A lot of money are* is the textbook case: of 13 raw plural-agreement hits, 10 are parse-shifted (people/parents/movies/times *are*; *a lot of money* in a relative clause), 1 is irrealis *were* (CGEL §3.5.1), 1 *'s* hit is contracted *has* (singular, matches prediction), and 1 is a genuine plural override.

The wrapper's KWIC mode is unreliable for API-driven retrieval from this script as of May 2026 (Playwright internal error on every attempted call), but that doesn't mean KWIC is unavailable for human use. Counts in the tables below were generated via list-mode queries; KWIC checks were then used to filter selected small cells where raw surface counts were likely to overstate the target construction.

A.2 A.2 Partitive vs. non-partitive *of*-complements

The contrast: partitive *of*-complements have definite complements (*of the money*, *of those books*); non-partitive *of*-complements have indefinite complements (*of money*, *of soldiers*, *of some paintings*).

A.2.1 Plural-only forms (*lots, bags, heaps, loads, oodles, stacks, piles*)

Non-partitive (X of N): *lots of people* 3,878; *lots of money* 1,376; *lots of time* 703; *loads of money* 109; *piles of money* 91; *loads of people* 90; *loads of time* 34; *heaps of money* 15; *heaps of people* 7; plus single-digit hits for several other combinations. Total: 6,311. KWIC checks of the single-digit cells show the expected mixture: *lots of day* is modifier use (*day-to-day*, *day hikes*, *day trips*); *heaps of time* and *piles of time* are genuine non-partitive mass-complement uses; *piles of people* is literal piling rather than the quantificational use.

Partitive (X of the N): *lots of the time* 7; *lots of the people* 5; *loads of the day* 1. Total: 13. *Piles* and *heaps* return 0 partitive hits with any complement tested.

Indefinite-NP complement (X of a N): *lots of a day* 0; *piles of a day* 0; *heaps of a day* 0; *loads of a day* 0; *lots/piles/heaps/loads of an afternoon* 0. Total: 0. Plural-only forms reject *of* [Det NP] whether the determiner is definite (*the*) or indefinite (*a/an*); their *of*-complements are bare/non-determined.

Partitive ratio: $13/(6{,}311 + 13) \approx 0.2\%$.

A.2.2 *Plenty*

Non-partitive: *plenty of time* 4,017; *plenty of people* 1,563; *plenty of money* 668; *plenty of day* 4 (likely modifier). Total: 6,252.

Partitive: *plenty of the time* 1; *plenty of the people* 1. Total: 2.

Partitive ratio: $2/6{,}254 \approx 0.03\%$.

A.2.3 *Rest, remainder*

Non-partitive (the X of N): *the rest of time* 75; *the rest of people* 12; *the rest of day* 7; *the remainder of time* 3; *the remainder of day* 2; *the rest of money* 1. Total: 100 raw. KWIC checks confirm that this raw total is conservative. *The rest of time* is mostly fixed or duration-level temporal use (*for the rest of time*, *spend the rest of time*), not the partitive-count contrast targeted here; *the rest of people* contains possessives and non-standard "other people" uses; *the rest of day* is temporal ellipsis or *day* as a modifier; and the *remainder* hits are technical or duration expressions.

Partitive (the X of the N): *the rest of the day* 1,932; *the rest of the time* 830; *the rest of the people* 272; *the rest of the money* 222; *the remainder of the day* 75; *the remainder of the time* 21; *the remainder of the money* 13. Total: 3,365.

Indefinite-NP complement: *the remainder of a day* 1.

Partitive (definite NP) ratio: $3{,}365/(3{,}365 + 100 + 1) \approx 97.1\%$ raw, higher after KWIC filtering of the non-target temporal, possessive, and generic uses in the raw non-partitive cells.

A.3 Override-direction agreement

Test of the "near-obligatory in central cases" claim. Subject NP plus finite verb agreement, by complement number/count.

Subject	Predicted	Plural agr.	Singular agr.	% predicted
a lot of people	plural	4,195 (ARE 3,406; WERE 789)	85 (IS 72; WAS 13)	98.0%
a lot of money	singular	1 (ARE, filtered)	90 (IS 61; WAS 29)	98.9%
a number of people	plural	100 (ARE 52; WERE 48)	0 (4 false positives)	100%
lots of people	plural	348 (ARE 288; WERE 60)	0 (6 false positives/non-standard)	100%
lots of money	singular	0 (1 false positive)	24 (IS 19; WAS 5)	100%
plenty of people	plural	79 (ARE 65; WERE 14)	0	100%
plenty of money	singular	0	6 (IS 4; WAS 2)	100%
the rest of the people	plural	19 (ARE 15; WERE 4)	0	100%

Subject	Predicted	Plural agr.	Singular agr.	% predicted
the rest of the money	singular	0	19 (IS 10; WAS 9)	100%

Override is in the predicted direction at 98–100% across all tested QN+complement combinations after filtering. *A lot of money* required filtering: of 13 raw plural-agreement hits, 10 were parse-shifted (the surface string *a lot of money are* appearing inside relative clauses modifying a plural-noun head, e.g., *people who earn a lot of money are successful*; *movies that make a lot of money are the biggest help*; *times when we... make a lot of money are gone*), 1 was the irrealis *were* (*if a lot of money were at stake*), and 1 *'s* hit was singular *has* (matches prediction); only 1 hit was a genuine plural override (*a lot of money are being raised*). Filtered count: 1 plural / 90 singular = 98.9% singular.

The other checked counter-direction cells are artefacts. *A number of people is/was* is headed by *condition*, *suggestion*, *ability*, or *counsel*, not by *number*; *lots of people is/was* is headed by expressions such as *murder*, *hauling*, or *disappearance*, with one non-standard existential hit; and *lots of money are* occurs inside a PP modifying plural *men*. The large *a lot of people is/was* cell remains unfiltered, so the 98.0% row is conservative.

A.4 A.4 *Bunch*: animate vs. inanimate aggregates

CGEL claims that *bunch* tilts collective for inanimate aggregates and transparent for groups of people. The COCA pilot supports the animate half but not the inanimate half.

Subject	Total tokens	*was*	*were*
Animate aggregates			
a bunch of people	1,421	0	28
a bunch of kids	448	0	12
a bunch of hooligans	12	0	0
Inanimate aggregates			
a bunch of flowers	68	0	0
a bunch of cars	–	0	1
a bunch of papers	–	0	0
a bunch of leaves	–	0	0
a bunch of things	–	0	1

Animates: 40 plural / 0 singular = 100% plural. KWIC checks found no parse-shift confound in these animate-plural hits. The CGEL "transparent for groups of people" claim is supported.

Inanimates: 2 plural / 0 singular. The CGEL "tilts collective for inanimate aggregates" claim isn't supported in COCA. Inanimate-bunch is mostly absent from subject position (typical use is as object: *she gave him a bunch of flowers*); the two inanimate-bunch hits in subject position both took plural rather than the CGEL-predicted singular. *A bunch of flowers was presented to the teacher* (the CGEL example) returns 0 hits in COCA.

B Compound transparency: Bell and Schäfer's empirical apparatus

This appendix carries the detailed exposition of Bell and Schäfer (2016) that §4 references in compressed form. The body keeps the headline finding (three predictors, scalar transparency, within-domain projection); the predicate inventory and stress speculation live here.

B.1 Levi's predicate inventory

Bell and Schäfer code the semantic relation between the constituents of each NN compound using Levi's (1978) nine predicates: CAUSE, HAVE, MAKE, USE, BE, IN, FOR, FROM, ABOUT (Bell & Schäfer, 2016, p. 167).

Three of the predicates (CAUSE, HAVE, MAKE) come in two variants depending on which constituent functions as the subject of the predicate:

- CAUSE1 *tear gas* ("gas that causes tears") versus CAUSE2 *drug deaths* ("deaths caused by drugs").
- HAVE1 *picture book* versus HAVE2 *government land*.
- MAKE1 *honeybee* versus MAKE2 *daisy chains*.

The other six predicates (USE, BE, IN, FOR, FROM, ABOUT) don't subdivide.

B.2 Metaphoric/metonymic shift coding

Bell and Schäfer's earlier work coded each compound and each constituent for metaphoric and metonymic shifts, treating shift as a binary feature (Bell & Schäfer, 2016, p. 167). The coding is independent of the predicate-relation assignment and contributes separately to predicting transparency ratings.

B.3 IN, FOR, and the stress hypothesis

Two of the Levi relations are positively associated with whole-compound transparency in Bell and Schäfer's models: IN and FOR. Drawing on Bell and Plag (2012), Bell and Schäfer suggest a possible mechanism: if more transparent constituents are less informative, and if compound stress falls on the more informative constituent, the IN/FOR effects on transparency would line up with the stress patterns observed for those compounds (Bell & Schäfer, 2016, p. 168). The connection is offered as a hypothesis that further work would need to test, not as an established mechanism.

B.4 N1/N2 asymmetry

Although whole-compound transparency is a function of the transparencies of the two constituents, N1 (the modifier) and N2 (the head) differ in the nature of their contribution. The asymmetry is one of the two headline conclusions of Bell and Schäfer's paper. The other is that all the significant predictors of rated transparency in their models are also independent predictors of processing speed, leading them to a tentative hypothesis: "perceived transparency may itself be a reflex of ease of processing" (Bell & Schäfer, 2016, 157, abstract). The hypothesis is offered as a direction for further work, not a result of the present study.

B.5 The Reddy et al. database

The empirical anchor of Bell and Schäfer's models is the database of Reddy and colleagues (2011), used as the dependent variable. The database consists of 30 transparency ratings for each of 90 two-part English NN compounds, totalling 8,100 ratings, collected via Amazon Mechanical Turk. Annotators rated, on a 0–5 scale, how literal each constituent was in the compound and how literal the compound as a whole was. Ratings are continuous; they don't partition compounds into transparent-versus-opaque classes (Bell & Schäfer, 2016, p. 164).

C Morphological priming: Heyer and Kornishova's design and statistical detail

This appendix carries the experimental detail from Heyer and Kornishova (2018) that §5 references in compressed form. The body keeps the design logic (short vs. long SOA; transparency continuous; key result that transparency modulates priming only at long SOAs); participant counts, exact SOAs, item-selection, and statistical results are here.

C.1 Participants and design

Two masked-priming experiments, one with English *-ness* nominalizations (49 native speakers: 24 in the short-SOA condition, 25 in the long-SOA condition) and one with Russian *-ost'* nominalizations (61 native speakers: 31 short-SOA, 30 long-SOA) (Heyer & Kornishova, 2018, p. 1115).

Targets in both experiments were the adjective bases (e.g., *pale, dark*). Primes came in three types:

- the corresponding nominalization as a related prime (e.g., *paleness* preceding *pale*);
- a simplex noun as an unrelated control prime;
- the adjective itself as an identity prime.

Each experiment used two stimulus-onset asynchronies (SOAs):

- short: 39 ms English, 33 ms Russian;
- long: 77 ms English, 67 ms Russian.

C.2 Items and transparency rating

Items spanned the full transparency range. At the opaque end, *business* (rated 3.17) is no longer interpretable as "state of being busy"; at the transparent end, *falseness* (rated 5.75) is fully derivable from *false*; intermediate items occupy the middle of the scale.

Out of 95 *-ness* items pre-tested, 30 were selected for the main experiment to cover the range continuously (Heyer & Kornishova, 2018, p. 1115). The Russian experiment used a parallel design with 30 *-ost'* nominalizations spanning a similar range.

Transparency was rated by separate groups of native speakers (28 English, 38 Russian) on a 1–7 Likert scale for semantic relatedness.

The two suffixes are highly comparable: both deadjectival, both producing abstract nouns denoting "state of being X" (e.g., *dark* + *ness* = 'darkness'; *bledn-* + *-ost'* = *blednost'* 'paleness'), and both productive (Heyer & Kornishova, 2018, p. 1115).

C.3 Statistical results

At short SOA in both languages, the data shows constant morphological priming across the transparency scale. The interaction between Prime Type and Transparency doesn't reach significance: priming effects are 37 ms in English and 16 ms in Russian, but they aren't modulated by how transparent the prime-target pair is rated (Heyer & Kornishova, 2018, pp. 1117–1119).

At long SOA, priming becomes transparency-modulated: more transparent prime-target pairs (e.g., *paleness-pale*) yield larger priming effects than less transparent pairs (e.g., *business-busy*). The three-way interaction between Prime Type, Transparency, and SOA is significant in English ($t = 2.10$, $p = 0.036$) and marginally so in Russian ($t = -1.69$, $p = 0.091$; the negative sign reflects factor coding rather than a reversed effect, and the direction of transparency-on-priming is consistent across the two languages). The English result establishes the SOA-dependent modulation; the Russian result trends in the same direction without reaching conventional significance (Heyer & Kornishova, 2018, pp. 1117–1119).

C.4 Rival parallel-processing position

The serial reading is contested. Feldman et al.'s (2015) *Must analysis of meaning follow analysis of form?* is the canonical parallel-immediate paper: a time-course analysis with a same-target design (each target compared after semantically similar and semantically dissimilar morphologically structured primes within subjects) and SOA treated as continuous from 34 to 100 ms. Their finding is that semantic similarity affects recognition latencies even at the shortest SOAs (34 and 48 ms), with the effect of shared semantics increasing linearly with SOA when long SOAs are intermixed. The serial form-then-meaning ordering predicts no semantic effect at SOAs in the 34–48 ms range; the parallel-immediate reading takes the observed effect as direct evidence against that prediction. Heyer and Kornishova's design (separate item sets at each SOA, transparency as a continuous predictor, two languages) gives the serial reading a different test: the absence of transparency modulation across the item set at the short SOA. The two designs probe different patterns. Same-target wedge (Feldman et al.) tests whether semantic similarity affects latencies on identical targets at very short SOAs; across-item gradient (Heyer and Kornishova) tests whether transparency rating predicts priming magnitude on a continuum. They aren't head-to-head, and the literature is mixed enough (the meta-table on p. 1121 of Heyer and Kornishova surveys 35 prior masked-priming studies) that the present paper takes the serial reading as Heyer and Kornishova's reading, not as a settled disciplinary verdict.

D A cross-modal probe design for the form–meaning family

This appendix sketches the cross-modal probe that §§7.2 and 7.4 flag as the open empirical question for the form–meaning family. The aim isn't a finished study but a tractable design whose outcome would settle the cross-domain projectibility question one way or the other.

D.1 Hypothesis and falsification

The form–meaning family of §§3–5 comprises three within-domain profiles: theoretical compositionality, rated compound transparency, time-sensitive masked priming. Each licenses within-domain projection. The cross-domain hypothesis is that the three are joint reflexes of a single underlying organization: shared representational structure that exposes form–meaning composability through different windows. The hypothesis is non-trivial because alternative organizations are available: each profile could be domain-internal, with no cross-domain alignment. The probe asks whether the four measurement windows align on the *same items*, which is the only test that distinguishes the cluster reading from the three-separate-profiles reading.

Falsification is built in. If corpus predictors, rated transparency, priming magnitude, and interpretation accuracy fail to align on shared items as joint consequences of representational structure, the family is three within-domain profiles and the cross-domain projectibility question resolves negatively. The diagnostic is a single, item-anchored study; sample-size and statistical-power planning would follow the four-window design rather than any single-window precedent.

D.2 Item set

Deadjectival nominalizations on *-ness* are the natural anchor: §5 establishes a within-domain transparency continuum with rated values from *business* (3.17) to *falseness* (5.75); the *-ness* items have been used in masked-priming work and are corpus-tractable; and parallel Russian *-ost'* items extend the design cross-linguistically without re-doing the architecture. A target set of 60–80 items spanning the rating continuum, with matched controls (frequency-matched simplex nouns, frequency-matched derivations on a different suffix), would support the four windows.

NN compounds (the §4 target) are an alternative or supplementary anchor. The same within-items logic applies, with Bell and Schäfer's (2016) corpus predictors providing the corpus measurement directly. Whether to run the probe on *-ness*, on NN compounds, or on both is a research-design choice; the probe's logical structure is identical across item types.

D.3 Measurements

Four within-items measurements are required.

- Corpus predictors. Frequency, productivity, and semantic-relation expectedness, computed per item from a balanced corpus (BNC, COCA, or COHA depending on the item type). Bell and Schäfer's measurement procedure for compounds extends naturally to nominalizations.
- Rated transparency. Native-speaker ratings on a 1–7 Likert scale, replicating Heyer and Kornishova's rating procedure with a separate participant group from the priming sub-study.
- Priming magnitude. Masked-priming reaction times at two SOAs (short, $\approx$30–40 ms; long, $\approx$60–80 ms), with the morphologically related prime, an unrelated simplex control, and an identity prime, replicating Heyer and Kornishova's design within the new item set.
- Interpretation accuracy. A paraphrase or sentence-completion task probing whether speakers can recover the relation between the derivative and its base in context. The task should be designed so that opaque items predict reduced accuracy and longer response times relative to transparent items.

The four windows probe representational structure at distinct stages of processing and explicit access. Their alignment, if found, is a non-trivial empirical pattern.

D.4 Statistical model

A joint analysis with item-level random effects and a latent representational-structure factor is the natural statistical move. Each measurement is a noisy reflex; the latent factor's loadings index how strongly each window reflects the underlying structure. The cluster hypothesis predicts non-trivial loadings on all four windows; the three-separate-profiles alternative predicts that the loadings partition by domain, with no shared factor.

The temporal asymmetry §7.4 flags between priming and interpretation is structural: priming probes early lexical access; interpretation probes explicit recovery; ratings are a metalinguistic judgement supplied at a separate, untimed task. The four windows aren't a temporal chain. The latent-factor model treats them as joint reflexes, with no claim that any one window causes any other.

D.5 Status

The probe doesn't exist. The form–meaning family is currently three within-domain profiles plus a research hypothesis. Section 7.4's Khalidi-refinement reading is conditional on the probe's outcome; absent the probe, Slater's SPC diagnostic is the relevant standard, and each within-domain profile is treated separately.

Acknowledgements

This paper was drafted with the assistance of large language models (Claude Opus 4.7 with the 1M-context configuration; OpenAI Codex 5.5). All content has been reviewed and revised by the authors, who take full responsibility for the final text.

References

Bell, M. J., & Schäfer, M. (2016). Modelling semantic transparency. *Morphology*, *26*(2), 157–199. https://doi.org/10.1007/s11525-016-9286-3

Boyd, R. (1991). Realism, anti-foundationalism and the enthusiasm for natural kinds. *Philosophical Studies*, *61*, 127–148. https://doi.org/10.1007/BF00385837

Boyd, R. (1999). Homeostasis, species, and higher taxa. In R. A. Wilson (Ed.), *Species: New interdisciplinary essays* (pp. 141–186). MIT Press. https://doi.org/10.7551/mitpress/6396.003.0012

da Cunha, Y., & Gibson, E. (2025). Syntactic complexity phenomena are better explained without empty elements mediating long-distance dependencies. *Cognitive Science*, *49*(8), e70088. https://doi.org/10.1111/cogs.70088

Feldman, L. B., Milin, P., Cho, K. W., Moscoso del Prado Martín, F., & O'Connor, P. A. (2015). Must analysis of meaning follow analysis of form? A time course analysis. *Frontiers in Human Neuroscience*, *9*, 111. https://doi.org/10.3389/fnhum.2015.00111

Gibson, E. (2025). *Syntax: A cognitive approach*. MIT Press.

Goldberg, A. E. (2006). *Constructions at work: The nature of generalization in language*. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780199268511.001.0001

Grosu, A. (2003). A unified theory of standard and transparent free relatives. *Natural Language and Linguistic Theory*, *21*(2), 247–331. https://doi.org/10.1023/A:1023387825622

Grosu, A. (2016). The semantics, syntax, and morphology of transparent free relatives revisited: A comparison of two approaches. *Natural Language and Linguistic Theory*, *34*(4), 1245–1280. https://doi.org/10.1007/s11049-015-9323-7

Haspelmath, M. (2010). Comparative concepts and descriptive categories in crosslinguistic studies. *Language*, *86*(3), 663–687. https://www.jstor.org/stable/40961695

Heyer, V., & Kornishova, D. (2018). Semantic transparency affects morphological priming ... eventually. *Quarterly Journal of Experimental Psychology*, *71*(5), 1112–1124. https://doi.org/10.1080/17470218.2017.1310915

Huddleston, R., & Pullum, G. K. (2002). *The Cambridge grammar of the English language*. Cambridge University Press.

Khalidi, M. A. (2018). Natural kinds as nodes in causal networks. *Synthese*, *195*(4), 1379–1396. https://doi.org/10.1007/s11229-015-0841-y

Kim, J.-B. (2026). *Form and function mismatches in English syntax: A construction-grammar perspective* (1st ed.). Routledge.

Kim, J.-B., & Reynolds, B. (2026, March). *Two kinds of transparency: Category flexibility and the verb inventory in TFRs* [Working note, in preparation.].

Newmeyer, F. J. (2007). Linguistic typology requires crosslinguistic formal categories. *Linguistic Typology*, *11*(1), 133–157. https://doi.org/10.1515/LINGTY.2007.012

Quine, W. V. (1956). Quantifiers and propositional attitudes. *The Journal of Philosophy*, *53*(5), 177–187. https://www.jstor.org/stable/2022451

Reynolds, B. (2026a). *The Homeostatic maintenance of English countability: Bidirectional inference and the stability of grammatical clusters* [Manuscript submitted to *Linguistics*.].

Reynolds, B. (2026b). *Words that won't hold still: How linguistic categories work* [Manuscript in preparation]. https://github.com/BrettRey/hpc-book

Schelfhout, C., Coppen, P.-A., & Oostdijk, N. (2003). Transparent free relatives. In S. Blaho, L. Vicente & M. de Vos (Eds.), *Proceedings of the twelfth conference of the student organization of linguistics in europe (CONSOLE XII)*.

Slater, M. H. (2015). Natural kindness. *British Journal for the Philosophy of Science*, *66*(2), 375–411. https://doi.org/10.1093/bjps/axt033

van Riemsdijk, H. (2006). Free relatives. In M. Everaert & H. van Riemsdijk (Eds.), *The Blackwell companion to syntax* (pp. 338–382). Blackwell.

Wilder, C. (1999). Transparent free relatives. In K. N. Shahin, S. Blake & E.-S. Kim (Eds.), *Proceedings of the west coast conference on formal linguistics* (pp. 685–689, Vol. 17). CSLI Publications.